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# WAY TO SUCCESS PUBLICATIONS

12th - Mathematics

## 12TH STD PTA MODEL QUESTION PAPER-I - ANSWER KEY 2020

### PART-I

1) c) I — creative
2) d) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$ — EX.1.8 (12)
3) a) 0 — EX.2.9 (1)
4) c) 1 — EX.2.9 (22)
5) c) n imaginary roots — EX.3.7 (3)
6) b) $\pi/3$ — EX. 4.6 (2)
7) c) $x = -17/4$ — creative
8) d) $-35 < m < 15$ — EX.5.6 (3)
9) a) $\lambda = -9/2$ — creative
10) b) 1 — EX.6.10 (19)
11) d) has no points of inflection — EX.7.10 (19)
12) b) $x = 2$ and $x = -2$ — creative
13) d) 4.8 cu cm — EX. 8.8 (7)
14) d) $2/27$ — EX. 9.10 (14)
15) b) 2 — creative
16) b) $P = ce^{-kt}$ — EX. 10.9 (23)
17) b) $\varphi(y/x) = kx$ — EX. 10.9 (17)
18) d) $l/2$, $l^2/12$ — Ex.11.6 (2)
19) d) $\binom{10}{5}\left(\frac{3}{5}\right)^5\left(\frac{2}{5}\right)^5$ — EX. 11.6 (12)
20) b) Z — EX. 12.3 (5)

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### PART-II

21. $AA^T = I$

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$A^T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

Example 1.11

$$AA^T = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\theta+\sin^2\theta & \cos\theta\sin\theta - \cos\theta\sin\theta \\ \cos\theta\sin\theta - \cos\theta\sin\theta & \sin^2\theta+\cos^2\theta \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$AA^T = I$

$\therefore$ A is orthogonal

22. $\frac{1-i}{3+i} + \frac{4i}{5} = \frac{1-i}{3+i} \times \frac{3-i}{3-i} + \frac{4i}{5}$

$= \frac{3-i-3i-1}{3^2-i^2} + \frac{4i}{5}$

$= \frac{2-4i}{9+1} + \frac{4i}{5}$

$= \frac{2(1-2i)}{10} + \frac{4i}{5}$

$= \frac{1-2i}{5} + \frac{4i}{5}$

$= \frac{1+2i}{5}$

$= \left|\frac{1+2i}{5}\right| = \frac{\sqrt{1^2+2^2}}{\sqrt{5^2}}$

$= \frac{\sqrt{5}}{5} = \frac{1}{\sqrt{5}}$

23. $\tan(\tan(-\pi/6))$

Ex 4.3 (11)

$= -\pi/6$, since $-\pi/6 \in (-\pi/2, \pi/2)$

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24. $3x^2+3y^2-12x+6y-9=0$

$\div 3$

$x^2+y^2-4x+2y-3=0$

$2g=-4 \mid 2f=2 \mid c=-3$

$g=-2 \quad f=1$

centre $(-g,-f)=(2,-1)$

Radius $r=\sqrt{g^2+f^2-c}$

$=\sqrt{2^2+1^2+3}$

$=\sqrt{4+1+3}$

$=\sqrt{8}$

$=2\sqrt{2}$

25. line $\frac{x-3}{-4}=\frac{y-4}{-7}=\frac{z+3}{12}$

Plane $5x-y+z=8$

$(x_1,y_1,z_1)=(3,4,-3)$

$(a,b,c)=(-4,-7,12)$

$(A,B,C)=(5,-1,1)$

$aA+bB+cC=(-4)(5)+(-7)(-1)+(12)(1)$

$=-20+7+12$

$=-8+7$

$=-1\neq 0$

So the normal to the plane is not perpendicular to the line. the given line doesnot lie in the plane

Ex 6.4 Q5

26. $\int_{-\log 2}^{\log 2} e^{-|x|}\,dx$

$f(x)=e^{-|x|}$

$f(-x)=e^{-|-x|}=e^{-|x|}=f(x)$

$f(x)$ is an even function

$\int_{-\log 2}^{\log 2} e^{-|x|}\,dx=2\int_{0}^{\log 2} e^{-|x|}\,dx$

$=2\left[e^{-|x|}\right]_0^{\log 2}$

$=2\left[e^{-\log 2}+e^{0}\right]$

$=2\left[-\frac{1}{2}+1\right]$

$=2\left[\frac{1}{2}\right]$

$=1$

Example 9.25

27. $\left(\frac{d^2y}{dx^2}\right)^2+\left(\frac{dy}{dx}\right)^2=x\sin\left(\frac{dy}{dx^2}\right)$

Ex. 10.1 1. (iii)

order 2

Degree does not exist

28. $f(x)$ is a probability mass function

$\Sigma f(x)=1$

$\frac{0^2+1}{k}+\frac{1^2+1}{k}+\frac{2^2+1}{k}=1$

$\frac{1}{k}+\frac{2}{k}+\frac{5}{k}=1$

$\frac{8}{k}=1$

$k=8$

Ex. 11.2 4 (i)

29. $a*b=a^b, \forall a,b\in N$

$(a*b)*c=a^b*c$

$=(a^b)^c$

Take $a=2, b=3, c=4$

$(a*b)*c=(2^3)^4=2^{12}\rightarrow(1)$

$a*b*c=a*b^c$

$=(a)^{b^c}$

Take $a=2, b=3, c=4$

$a*(b*c)=(2)^{3^4}$

$=2^{81}\rightarrow(2)$

From (1), (2)

$(a*b)*c\neq a*(b*c)$

Example 12.6 (3)

30. $\lim_{x\to 0}\frac{xe^x-\sin x}{x}=\frac{0e^x-\sin 0}{0}$

$=\frac{0}{0}$

(∵ Indeterminant form)

Apply 'L' Hospital rule

$\lim_{x\to 0}\frac{xe^x-\sin x}{x}=\lim_{x\to 0}\frac{xe^x+e^x-\cos x}{1}$

$=\lim_{x\to 0} xe^x+e^x-\cos x$

$=0e^0+e^0-\cos 0$

$=0+1-1$

$=0$

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PART - III

31. $A = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$

$A \sim \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -3 & 1 & -2 \\ 0 & -6 & 2 & -4 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 5R_1 \end{matrix}$

$\sim \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -3 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_3 \to R_3 - 2R_2$

$P(A) = 2$

Ex. 1.2 - 2 (i)

32. $z^3 + 2\bar{z} = 0$

$z^3 = -2\bar{z}$

$|z|^3 = |-2|\,|\bar{z}|$

$|z|^3 - 2|\bar{z}| = 0$

$|z|[|z|^2 - 2] = 0$

$z = 0$ or $|z|^2 - 2 = 0$

$|z|^2 = 2$

$z\bar{z} = 2$

$\bar{z} = \frac{2}{z}$

$z^3 + 2\bar{z} = z^3 + 2\left(\frac{2}{z}\right) = 0$

$\frac{z^4 + 4}{z} = 0$

$z^4 + 4 = 0$

$z = 0$, $z^4 + 4 = 0$ has five solution

Ex. 2.5 - (9)

33. Sum of the polynomial equation

$= 1 - 5 - 4 + 20$

$= 21 - 9$

$= 12 \neq 0$

$(x-1)$ is not a factor

Example 3.25

| | | | | |
|---|---|---|---|---|
| 2 | 1 | -5 | -4 | -20 |
| | 0 | 2 | -6 | -20 |
| | 1 | -3 | -10 | 0 |

$x^3 - 3x - 10 = 0$

$(x-5)(x+2) = 0 \Rightarrow x = 5, -2$

solution $x = 2, 5, -2$

34. $3x^2 + (3-p)xy + 2y^2 - 2px = 8pq \to (1)$

coefficient of $xy = 0$

$(3-p) = 0$

$-p = -3 \Rightarrow \boxed{p = 3}$

coefficient of $x^2$ = coefficient of $y^2$

$3 = 2$

Sub $p = 3$, $q = 3$ in (1)

$3x^2 + (3-3)xy + 3y^2 - 6x = 8(3)(3)$

$3x^2 + 3y^2 - 6x - 72 = 0$

$\div 3$

$x^2 + y^2 - 2x - 24 = 0$

$2g = -2 \mid 2f = 0 \mid c = -24$

$g = -1 \mid f = 0$

centre $(-g, -f) = (1, 0)$

Radius $r = \sqrt{g^2 + f^2 - c}$

$= \sqrt{1 + 0 + 24}$

$= \sqrt{25} = 5$

$\boxed{r = 5}$

Ex. 5.1 - (12)

35. $y = x^3 - 6x^2 + x + 3 \to (1)$

$x + y = 1729 \to (2)$

Diff (1) with r. to 'x'

$\frac{dy}{dx} = 3x^2 - 12x + 1$

= slope of tangent

slope of normal $= \frac{-1}{\text{slope of tangent}}$

$= \frac{-1}{3x^2 - 12x + 1}$

Diff (2) w. r. to 'x'

$1 + \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -1$

$\therefore m_1 = m_2$

$\frac{-1}{3x^2 - 12x + 1} = -1$

Ex. 7.2 - (3)

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$3x^2 - 12x = 0$

$3x(x-4) = 0$

$x = 0, \quad x = 4$

Sub $x = 0$ in (1)

$y = 0 - 0 + 0 + 3 = 3$

$\boxed{y = 3}$

Sub $x = 4$ in (1)

$y = 64 - 6(16) + 4 + 3$

$= 64 - 96 + 4 + 3$

$= 71 - 96$

$\boxed{y = -25}$

The points $(0, 3)$, $(4, -25)$

36. $u(x, y) = \dfrac{x^2 + y^2}{\sqrt{x+y}}$

$u(\lambda x, \lambda y) = \dfrac{\lambda^2 x^2 + \lambda^2 y^2}{\sqrt{\lambda x + \lambda y}}$

$= \dfrac{\lambda^2 [x^2 + y^2]}{\sqrt{\lambda}\,(\sqrt{x+y})}$

$= \dfrac{\lambda^2 [x^2 + y^2]}{\lambda^{1/2} [\sqrt{x+y}]}$

$= \lambda^{2 - 1/2} \cdot \dfrac{x^2 + y^2}{\sqrt{x+y}}$

$= \lambda^{3/2}\, u(x, y)$

Hence $u$ is a homogeneous function of degree $\frac{3}{2}$

By Euler's theorem $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{3}{2}u$

EX. 8.4 (4)

37. $\int_0^1 \dfrac{2x}{1+x^2}\,dx$

$= [\log(1+x^2)]_0^1$

$= \log(1+1^2) - \log(1+0)$

$= \log 2 - 0$

$= \log 2$

38. $\cos x \cos y\, dy - \sin x \sin y\, dx = 0$

$\cos x \cos y\, dy = \sin x \sin y\, dx$

$\dfrac{\cos y}{\sin y}\, dy = \dfrac{\sin x}{\cos x}\, dx$

$\int \cot y\, dy = \int \tan x\, dx$

$\log \sin y = \log \sec x + \log c$

$\sin y = c \sec x$

$\cot x \sin y = C$

EX. 11.4 (10)

39. $E(X+3) = 10$

$E(X) = 10 - 3$

$E(X) = 7$

$E(X+3)^2 = 116$

$E(X^2) + E(6X) + E(9) = 116$

$E(X^2) + 6E(X) = 116 - 9$

$E(X^2) + 42 = 107$

$E(X^2) = 107 - 42$

$E(X^2) = 65$

$\text{Var}(X) = E(X^2) - [E(X)]^2$

$= 65 - 49$

$= 16$

$\sigma^2 = 16, \quad \mu = 7$

40. $[\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}]$

$= \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} [\vec{a}, \vec{b}, \vec{c}]$

$= [1(1-0) - 1(0-1) + 0(0-1)]\,[\vec{a}\ \vec{b}\ \vec{c}]$

$= [1 + 1]\,[\vec{a}\ \vec{b}\ \vec{c}]$

$= 2\,[\vec{a}\ \vec{b}\ \vec{c}]$

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Part - IV

41. (a) $A=\begin{bmatrix}4&3&6\\1&5&7\\2&9&1\end{bmatrix}, B=\begin{bmatrix}25\\13\\1\end{bmatrix}, X=\begin{bmatrix}x\\y\\z\end{bmatrix}$

$$[A/B]=\left[\begin{array}{ccc|c}4&3&6&25\\1&5&7&13\\2&9&1&1\end{array}\right]$$

$$=\left[\begin{array}{ccc|c}1&5&7&13\\4&3&6&25\\2&9&1&1\end{array}\right] R_1 \leftrightarrow R_2$$

$$=\left[\begin{array}{ccc|c}1&5&7&13\\0&-17&-22&-27\\0&-1&-13&-25\end{array}\right] \begin{matrix}R_2\to R_2-4R_1\\R_3\to R_3-2R_1\end{matrix}$$

$$=\left[\begin{array}{ccc|c}1&5&7&13\\0&17&22&27\\0&1&13&25\end{array}\right] \begin{matrix}R_2\to -R_2\\R_3\to -R_3\end{matrix}$$

$$=\left[\begin{array}{ccc|c}1&5&7&13\\0&17&22&27\\0&0&199&398\end{array}\right] R_3\to 17R_3-R_2$$

$\rho(A)=3$, $\rho(A/B)=3$

$\rho(A)=\rho(A/B)=3=$ no. of unknowns

∴ The system has unique solution

$x+5y+7z=13 \to (1)$

$17y+22z=27 \to (2)$

$199z=398 \to (3)$

$$z=\frac{398}{199}=2$$

$\boxed{z=2}$

sub $z=2$ in (2)

$17y+22(2)=27$

$17y+44=27$

$17y=-17$

$\boxed{y=-1}$

sub $y=-1, z=2$ in (1)

$x+5(-1)+7(2)=13$

$x-5+14=13$

$\boxed{x=4}$

41. (b) $\frac{z-i}{z+2}=\frac{x+iy-i}{x+iy+2}$

$$=\frac{x+i(y-1)}{(x+2)+iy}\times\frac{(x+2)-iy}{(x+2)-iy}$$

$$=\frac{x(x+2)-ixy+i(y-1)(x+2)-i(y-1)y}{(x+2)^2+y^2}$$

$$=\frac{x(x+2)+y(y-1)+i(y-1)(x+2)-ixy}{(x+2)^2+y^2}$$

$\arg\left[\frac{z-i}{z+2}\right]=\pi/4$

$\tan\left(\frac{(y-1)(x+2)-xy}{x(x+2)+y(y-1)}\right)=\pi/4$

Ex. 2.4 (6)

$\frac{(y-1)(x+2)-xy}{x(x+2)+y(y-1)}=\tan\pi/4$

$\frac{(y-1)(x+2)-xy}{x(x+2)+y(y-1)}=1$

$(y-1)(x+2)-xy=x(x+2)+y(y-1)$

$xy+2y-x-2-xy=x^2+2x+y^2-y$

$-x+y+2y-2x-x-y^2-2=0$

$-x^2-y^2-5x+3y-2=0$

× by −1

$x^2+y^2+3x-3y+2=0$

42. (a)

$6x^4-35x^3+62x^2-35x+6=0$

$6x^2\left(x^2+\frac{1}{x^2}\right)-35x^2\left(x+\frac{1}{x}\right)+62x^2=0$

$x^2\left[6\left(x^2+\frac{1}{x^2}\right)-35\left(x+\frac{1}{x}\right)+62\right]=0$

$6\left(x^2+\frac{1}{x^2}\right)-35\left(x+\frac{1}{x}\right)+62=0$

Ex. 2.5 (1)

Let $x+\frac{1}{x}=y$

squaring on both side

$\left(x+\frac{1}{x}\right)^2=y^2$

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$x^2+\frac{1}{x^2}+2=y^2$

$x^2+\frac{1}{x^2}=y^2-2$

$\therefore 6(y^2-2)-35y+62=0$

$6y^2-35y+50=0$

$(3y-10)(3y-5)=0$

$y=\frac{10}{3}, y=\frac{5}{2}$

$x+\frac{1}{x}=y$

$\frac{x^2+1}{x}=y$

$x^2+1=yx$

| Put $y=\frac{10}{3}$ | Put $y=5/2$ |
|---|---|
| $x^2+1=\frac{10}{3}x$ | $x^2+1=5/2\,x$ |
| $3x^2+3=10x$ | $2x^2+2=5x$ |
| $3x^2-10x+3=0$ | $2x^2-5x+2=0$ |
| $(x-3)(x-1/3)$ | $(x-2)(x-1/2)=0$ |
| $x=3, x=1/3$ | $x=2, x=1/2$ |

The roots are $3, 1/3, 2, 1/2$

42. b) Equation of Parabola

$(x-h)^2=-4a(y-k)$

Vertex $(0.5, 4)$

$(x-0.5)^2=-4a(y-4) \rightarrow (1)$

$(0,0)$ is a point of Parabola

$(0-0.5)^2=-4a(0-4)$

$(-\frac{1}{2})^2=16a$

$\frac{1}{4}=16a$

$a=\frac{1}{64}$

Ex. 5.5 (20)

sub (1)

$(x-0.5)^2=-4\left(\frac{1}{64}\right)(y-4)$

$(x-0.5)^2=-\frac{1}{16}(y-4)$

$(0.75, y_1)$ is a point on Parabola

$(0.75-0.5)^2=-\frac{1}{16}(y_1-4)$

$\left(\frac{3}{4}-\frac{1}{2}\right)^2=-\frac{1}{16}(y_1-4)$

$\left(\frac{3-2}{4}\right)^2=-\frac{1}{16}(y_1-4)$

$\frac{1}{16}=-\frac{1}{16}(y_1-4)$

$1=-y_1+4$

$-y_1=1-4$

$y_1=3$

Required distance 3m

43. a) $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&0\\1&-1&-4\end{vmatrix}=4\hat{i}+4\hat{j}$

$\vec{c}\times\vec{d}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\0&3&-1\\2&5&1\end{vmatrix}=8\hat{i}-2\hat{j}-6\hat{k}$

Example 6.23 (ii)

$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\4&4&0\\8&-2&-6\end{vmatrix}$

$=-24\hat{i}+24\hat{j}-40\hat{k} \rightarrow (1)$

$[\vec{a}\ \vec{b}\ \vec{d}]\vec{c}=[(\vec{a}\times\vec{b})\cdot\vec{d}]\vec{c}$

$=28(3\hat{j}-\hat{k})$

$=84\hat{j}-28\hat{k}$

$[\vec{a}\ \vec{b}\ \vec{c}]\vec{d}=[(\vec{a}\times\vec{b})\cdot\vec{c}]\vec{d}$

$=24\hat{i}+6\hat{j}+12\hat{k}$

$[\vec{a}\ \vec{b}\ \vec{d}]\vec{c}-[\vec{a}\ \vec{b}\ \vec{c}]\vec{d}$

$=84\hat{j}-28\hat{k}-24\hat{i}-6\hat{j}-12\hat{k}$

$=-24\hat{i}+24\hat{j}-40\hat{k} \rightarrow (2)$

From (1), (2)

$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})=[\vec{a}\ \vec{b}\ \vec{d}]\vec{c}-[\vec{a}\ \vec{b}\ \vec{c}]\vec{d}$

43. b)

$\vec{r}=6\hat{i}-\hat{j}+\hat{k}+s(-\hat{i}+2\hat{j}+\hat{k})+t(-5\hat{i}-\hat{j}-5\hat{k})$

Ex. 6.7 (4)

$\vec{a}=6\hat{i}-\hat{j}+\hat{k}$, $\vec{b}=-\hat{i}+2\hat{j}+\hat{k}$, $\vec{c}=-5\hat{i}-\hat{j}-5\hat{k}$

$(\vec{r}-\vec{a})\cdot(\vec{b}\times\vec{c})=0$

$\vec{b}\times\vec{c}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-1&2&1\\-5&-1&-5\end{vmatrix}$

$=-6\hat{i}-10\hat{j}+14\hat{k}$

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$(\vec{r} - 6\hat{i} - \hat{j} + \hat{k}) \cdot (-6\hat{i} - 10\hat{j} + 14\hat{k}) = 0$

$\vec{r} \cdot (3\hat{i} + 5\hat{j} - 7\hat{k}) = 6$

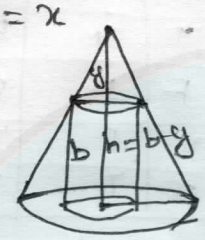
$r = x\hat{i} + y\hat{j} + z\hat{k}$

$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (3\hat{i} + 5\hat{j} - 7\hat{k}) = 6$

$3x + 5y - 7z = 6$

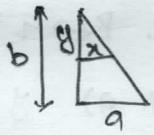
$3x + 5y - 7z - 6 = 0$

44. a) Radius of cylinder $= x$




$\frac{y}{b} = \frac{x}{a}$

$y = \frac{bx}{a} \rightarrow (1)$




Volume $V = \pi r^2 h$

$V = \pi\left(bx^2 - \frac{bx^3}{a}\right)$ | $r^2$

$\frac{dV}{dx} = \pi\left(2bx - \frac{3bx^2}{a}\right)$

$\frac{d^2V}{dx^2} = \pi\left(2b - \frac{6bx}{a}\right)$

EX. 7.9 - (2)

$\frac{dV}{dx} = 0 \Rightarrow \pi\left(2bx - \frac{3bx^2}{a}\right) = 0$

$x = \frac{2a}{3}$

Let $x = \frac{2a}{3} \Rightarrow \frac{d^2V}{dx^2} = \pi\left(2b - \frac{6b}{a}\left(\frac{2a}{3}\right)\right)$

$= \pi(2b - 4b)$

$= \pi(-2b) < 0$

It is maximum

Maximum volume of cylinder

$V = \pi\left(b\left(\frac{2a}{3}\right)^2 - \frac{b}{a}\left(\frac{2a}{3}\right)^3\right)$

$= \frac{4}{9} \times \frac{1}{3}\pi a^2 b$

$= \frac{4}{9} \times$ Volume of Cone

44. b) $f(x) = \log(1+x)$, $f(0) = \log 1 = 0$

$f'(x) = \frac{1}{1+x}$, $f'(0) = \frac{1}{1+0} = 1$

$f''(x) = \frac{-1}{(1+x)^2}$, $f''(0) = \frac{-1}{(1+0)} = -1$

$f'''(x) = \frac{2}{(1+x)^3}$, $f'''(0) = \frac{2}{(1+0)^3} = 2$

$f^{IV}(x) = \frac{-6}{(1+x)^3}$, $f^{IV}(0) = \frac{-6}{(1+0)^4} = -6$

Example 7.20

$f(x) = f(0) + \frac{f'(0)}{1!}x^2 + \frac{f''(0)}{2!}x^2 + \cdots$

$\log(1+x) = 0 + \frac{2}{1!}x - \frac{1}{2!}x^2 + \frac{2}{3!}x^3 \cdots$

$= x - \frac{1}{2}x^2 + \frac{x^3}{3} \cdots ; -1 < x \leq 1$

45. a)

$\frac{\partial u}{\partial x} = yz$ | $\frac{\partial u}{\partial y} = xz$ | $\frac{\partial u}{\partial z} = xy$

$\frac{dx}{dt} = -e^{-t}$ | $\frac{dy}{dt} = e^{-t}\cos t + \sin t(e^{-t})$ | $\frac{dz}{dt} = \cos t$

$\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}$

$= yz(-e^{-t}) + xz(e^{-t}\cos t + \sin t\, e^{-t}) + xy(\cos t)$

$= e^{-t}\cos t(\sin t)(-e^{-t}) + e^{-t}\sin t(e^{-t}\cos t - \sin t\, e^{-t}) + (e^{-t})(e^{-t}\cos t)\cos t$

EX. 8.6 - (4)

$= -e^{-2t}\sin^2 t + e^{-2t}(\sin t(\cos t - \sin t)) + e^{-2t}(\sin t\cos t)$

$= e^{-2t}[-\sin^2 t + \sin t\cos t - \sin^2 t + \sin t\cos t]$

$= e^{-2t}[-2\sin^2 t + 2\sin t\cos t]$

$= -e^{-2t}[2\sin^2 t - 2\sin t\cos t]$

$= e^{-2t}[2\sin^2 t - \sin 2t]$

$= -e^{-2t}[\cos t\sin t + \sin^2 t + \sin t\cos t - \cos^2 t]$

$= -e^{-2t}[\sin 2t - \cos 2t]$

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45. b) $(y - e^{\sin^{-1}x})\frac{dx}{dy} = -\sqrt{1-x^2}$

$\frac{dy}{dx}\sqrt{1-x^2} + y = e^{\sin^{-1}x}$

÷ by $\sqrt{1-x^2}$

$\frac{dy}{dx} + \frac{y}{\sqrt{1-x^2}} = \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}$

$P = \frac{1}{\sqrt{1-x^2}}$, $Q = \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}$

$e^{\int P dx} = e^{\int \frac{1}{\sqrt{1-x^2}} dx}$

$= e^{\sin^{-1}x}$

$y e^{\int P dx} = \int Q e^{\int P dx} dx + c$

$y e^{\sin^{-1}x} = \int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}} e^{\sin^{-1}x} dx + c$

$\sin^{-1}x = t$

$\frac{1}{\sqrt{1-x^2}} dx = dt$

$= \int e^t . e^t dt + c$

$= \int e^{2t} dt + c$

$= \frac{e^{2t}}{2} + c$

$y e^{\sin^{-1}x} = \frac{e^{2\sin^{-1}x}}{2} + c$

Ex. 10.7 (11)

46. a) (i) Binomial distribution

$x$ denotes the number of success

$x = 0, 1, 2 \ldots 10$

$P = 1/4$, $q = 1 - P = 1 - 1/4 = 3/4$

$P(x = x) = {}^{10}C_x \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{10-x}$

(ii) $P(x = 7) = {}^{10}C_7 \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^{10-7}$

$= {}^{10}C_7 \left(\frac{1}{4^7}\right)\left(\frac{3^3}{4^3}\right)$

$= 120 \left(\frac{3^3}{4^{10}}\right)$

(iii) $P(x \geq 1) = 1 - P(x < 1)$

$= 1 - P(x = 0)$

$= 1 - {}^{10}C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{10-0}$

$= 1 - \left(\frac{3}{4}\right)^{10}$

46. b) $P \leftrightarrow q$

| P | q | $P \leftrightarrow q$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | F |
| F | F | T |

$(P \wedge q) \vee (\neg P \wedge \neg q)$

| P | q | $P \wedge q$ | $\neg P$ | $\neg q$ | $\neg P \wedge \neg q$ | $(P \wedge q) \vee (\neg P \wedge \neg q)$ |
|---|---|---|---|---|---|---|
| T | T | T | F | F | F | T |
| T | F | F | F | T | F | F |
| F | T | F | T | F | F | F |
| F | F | T | T | T | T | T |

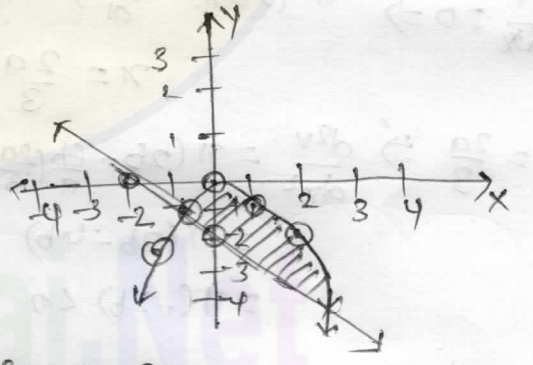
$\therefore P \leftrightarrow q \equiv (P \wedge q) \vee (\neg P \wedge \neg q)$

47. a) $y = x^2$

| x | 0 | +1 | −1 | 2 | −2 |
|---|---|---|---|---|---|
| y | 0 | 1 | 1 | 4 | 4 |

$x + y + 2 = 0$

| x | 0 | −1 | −2 | −3 | 2 |
|---|---|---|---|---|---|
| y | −2 | −1 | 0 | 1 | −4 |




$x^2 = -x - 2$

$x^2 - x - 2 = 0$

$(x+1)(x-2) = 0 \Rightarrow x = -1, x = 2$

$\text{Area} = \int_a^b (y_1 - y_2)\, dx$

$= \int_{-1}^{2} (-x^2 - (x-2))\, dx$

$= \int_{-1}^{2} (-x^2 + x + 2)\, dx$

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$$= \left[\frac{-x^3}{3} + \frac{x^2}{2} + 2x\right]_{-1}^{2}$$

$$= \left[\left(-\frac{8}{3} + \frac{4}{2} + 4\right) - \left(\frac{1}{3} + \frac{1}{2} - 2\right)\right]$$

$$= \left(\frac{-16 + 12 + 24}{6}\right) - \left(\frac{2 + 3 - 12}{6}\right)$$

$$= \frac{20}{6} - \left(-\frac{7}{6}\right)$$

$$= \frac{27}{6} = \frac{9}{2}$$

47. b)

Example 4.22

$\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$

$\cos^{-1}x = \alpha$, $\cos^{-1}y = \beta$

$\alpha + \beta + \cos^{-1}z = \pi$

$\alpha + \beta = (\pi - \cos^{-1}z)$

$\cos(\alpha+\beta) = \cos(\pi - \cos^{-1}z)$

$\cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha+\beta)$

$xy - \sqrt{1-x^2}\sqrt{1-y^2} = \cos(\alpha+\beta)$

$xy - \sqrt{1-x^2}\sqrt{1-y^2} = z$

$(xy + z) = \sqrt{1-x^2}\sqrt{1-y^2}$

squaring on both side

$(xy+z)^2 = \left(\sqrt{1-x^2}\right)\left(\sqrt{1-y^2}\right)^2$

$x^2 + y^2 + z^2 + 2xyz = 1$

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## PTA Model Question Paper – 2 Answer key (EM) 2019 - 2020

### Part – I

| Answer all the questions | |
|---|---|
| 1. | d) $2A^{-1}$ |
| 2. | a) $\begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$ |
| 3. | a) $\frac{1}{2}\lvert z\rvert^2$ |
| 4. | a) real axis |
| 5. | b) $\frac{3\pi}{4}$ |
| 6. | a) $[0,\pi]\backslash\left\{\frac{\pi}{2}\right\}$ |
| 7. | c) 10 |
| 8. | b) $x=-1$ |
| 9. | c) an ellipse |
| 10. | c) 45° |
| 11. | c) $200\pi$ |
| 12. | c) $\frac{\pi}{2}$ |
| 13. | c) $yx^{y-1}$ |
| 14. | d) 2 |
| 15. | c) $\frac{28}{3}$ |
| 16. | b) $-2$ |
| 17. | d) 2 |
| 18. | b) 2 |
| 19. | d) $(\neg p \wedge \neg q) \rightarrow (\neg p \vee \neg q)$ |
| 20. | d) $\sqrt{5}$ is an irrational number |

### Part – II

21. TB : 64 (Example : 2.7)
22. CQ
23. CQ
24. TB : 198 (Ex. 5.2 – 1 (iv))
25. TB : 5 (Example 7.5)
26. CQ
27. TB : 154 (Ex. 10.3 - 8)
28. TB : 247 (Example : 12.17)
29. TB : 235 (Ex. 12.1 – 3)
30. CQ

### Part – III

31. TB : 92 (Ex. 2.8 - 7)
32. TB : 127 (Ex. 3.6 - 3)
33. TB : 197 (Ex. 5.2 - 6)
34. TB : 237 (Ex. 6.2 - 5)
35. CQ
36. TB : 21 (Ex. 7.3 – 5 (ii))
37. TB : 118 (Example : 9.37)
38. TB : 152 (Ex. 10.2 - 2)
39. CQ
40. CQ

## PTA Model Question Paper – 2 Answer key (TM) 2019 - 2020

### பகுதி – I

| அனைத்து வினாக்களுக்கும் விடையளி | |
|---|---|
| 1. | d) $2A^{-1}$ |
| 2. | a) $\begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$ |
| 3. | a) $\frac{1}{2}\lvert z\rvert^2$ |
| 4. | a) மெய் அச்சின் மீது |
| 5. | b) $\frac{3\pi}{4}$ |
| 6. | a) $[0,\pi]\backslash\left\{\frac{\pi}{2}\right\}$ |
| 7. | c) 10 |
| 8. | b) $x=-1$ |
| 9. | c) நீள்வட்டம் |
| 10. | c) 45° |
| 11. | c) $200\pi$ |
| 12. | c) $\frac{\pi}{2}$ |
| 13. | c) $yx^{y-1}$ |
| 14. | d) 2 |
| 15. | c) $\frac{28}{3}$ |
| 16. | b) $-2$ |
| 17. | d) 2 |
| 18. | b) 2 |
| 19. | d) $(\neg p \wedge \neg q) \rightarrow (\neg p \vee \neg q)$ |
| 20. | d) $\sqrt{5}$ ஒரு விகிதமுறா எண் |

### பகுதி - II

21. TB : 70 (எகா. 2.7)
22. CQ
23. CQ
24. TB : 214 (பயிற்சி 5.2 – 1 (iv))
25. TB : 5 (எகா. 7.5)
26. CQ
27. TB : 163 (பயிற்சி 10.3 - 8)
28. TB : 263 (எகா. 12.17)
29. TB : 250 (பயிற்சி 12.1 – 3)
30. CQ

### பகுதி – III

31. TB : 98 (பயிற்சி 2.8 - 7)
32. TB : 139 (பயிற்சி 3.6 - 3)
33. TB : 215 (பயிற்சி 5.2 - 6)
34. TB : 259 (பயிற்சி 6.2 - 5)
35. CQ
36. TB : 22 (பயிற்சி 7.3 – 5 (ii))
37. TB : 123 (எகா. 9.37)
38. TB : 159 (பயிற்சி 10.2 - 2)
39. CQ
40. CQ



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| Part – IV | பகுதி – III |
|---|---|
| 41. a) TB : 41 (Example : 1.34)<br>b) TB : 106 (Ex. 3.1 - 5)<br>42. a) TB : 162 (Example: 4.20)<br>b) CQ<br>43. a) TB : 263 (Example: 6.7 - 4)<br>b) TB : 255 (Example: 6.5 - 7 )<br>44. a) TB : 44 (Ex. 7.7 – 1 (iii))<br>b) TB : 119 (Example: 9.40)<br>45. a) TB : 130 (Example: 9.55)<br>b) TB : 174 (Ex. 10.8 - 3)<br>46. a) TB : 202 (Ex. 11.3 - 4)<br>b) TB : 249 (Ex. 12.2 – 7 (iii))<br>47. a) CQ<br>b) CQ | 41. a) TB : 46 (எகா. 1.34)<br>b) TB : 115 (பயிற்சி 3.1 - 5)<br>42. a) TB : 179 (எகா. 4.20)<br>b) CQ<br>43. a) TB : 289 (எகா. 6.7 - 4)<br>b) TB : 279 (எகா. 6.5 - 7 )<br>44. a) TB : 46 (பயிற்சி 7.7 – 1 (iii))<br>b) TB : 124 (எகா. 9.40)<br>45. a) TB : 135 (எகா. 9.55)<br>b) TB : 184 (பயிற்சி 10.8 - 3)<br>46. a) TB : 214 (பயிற்சி 11.3 - 4)<br>b) TB : 265 (பயிற்சி 12.2 – 7 (iii))<br>47. a) CQ<br>b) CQ |

CQ - Creative Question



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## Answers

### Part - I

1, (d) $2A^{-1}$

2, (a) $adj(adjA) = A = \begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$

3, (a) $\frac{1}{2}|z|^2$

4, (a) Real axis.

5, (b) $\frac{3\pi}{4}$

6, (a) $[0\ \pi] \setminus \{\frac{\pi}{2}\}$.

7, (c) 10

8, (b) $x = -1$

9, (c) an ellipse.

10, (c) 45°

11, (c) $200\pi$

12, (c) $\frac{\pi}{2}$

13, (c) $y x^{y-1}$

14, (d) 2

15, (c) $\frac{28}{3}$

16, (b) $-2$

17, (d) 2

18, (b) 2

19, (d) $(\neg p \wedge \neg q) \rightarrow (\neg p \vee \neg q)$ is

20, (d) $\sqrt{5}$ is an irrational number.

### Part - II

21, $z = (2+3i)(1-i) = 5+i$

$z^{-1} = \frac{1}{z} = \frac{1}{5+i}$

$= \frac{1}{5+i} \times \frac{5-i}{5-i} = \frac{5}{26} - \frac{i}{26}$

22, $z = -6+8i$; $|z| = 10$

$\sqrt{-6+8i} = \pm\left(\sqrt{\frac{10-6}{2}} + i(1)\sqrt{\frac{10+6}{2}}\right)$

$= \pm(\sqrt{2} + i2\sqrt{2})$

$(OR) = (2-8) + i\,2(\sqrt{2})(2\sqrt{2})$

$-6+8i = [(\sqrt{2})^2 - (2\sqrt{2})^2] + i\,2(\sqrt{2})(2\sqrt{2})$

$= (\sqrt{2} + i2\sqrt{2})^2$

$\therefore \sqrt{-6+8i} = \pm(\sqrt{2} + i2\sqrt{2})$

23, $\operatorname{cosec}^{-1}(-\sqrt{2}) = \sin^{-1}(-\frac{1}{\sqrt{2}})$

$= \sin^{-1}(\sin(-\pi/4))$

$= -\frac{\pi}{4}$.

$\therefore$ Principal value of $\operatorname{cosec}^{-1}(-\sqrt{2})$ is $-\frac{\pi}{4}$.

24, End of Latus Rectum $L(a, 2a)$ $L'(a, -2a)$

$= L(4, 8)$ $L'(4, -8)$

$a = 4$.

Centre $= C(0, 0)$

Eqn. of Parabola

$y^2 = 4ax$

$= 4(4)x$

$y^2 = 16x$

25, $S = 128t - 16t^2$

$v = \frac{ds}{dt} = 128 - 32t$.

Put $v = 0 \Rightarrow 128 - 32t = 0$

$4 - t = 0$

$t = 4$

Max. Height $= S = 128(4) - 16(16)$

$= 512 - 256$

$= 256$ ft

26, $f_x = 3x^2 - 6x + 5$

$f_x$ at $(1, -2)$: $3 - 6 + 5 = 8 - 6$

$= 2$.

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27) $xy = ae^x + be^{-x} + x^2 \rightarrow (1)$

$x y_1 + y(1) = ae^x + b(-e^{-x}) + 2x$

$xy_1 + y = ae^x - be^{-x} + 2x$

Again diff w.r.t $x$

$xy_2 + y_1 + y_1 = ae^x - b(-e^{-x}) + 2$

$xy_2 + 2y_1 = ae^x + be^{-x} + 2$

From (1) $xy - x^2 = ae^x + be^{-x}$.

∴ D.E is $xy_2 + 2y_1 = (xy - x^2) + 2$

$\therefore xy_2 + 2y_1 + x^2 - xy - 2 = 0$

28)

| $p$ | $q$ | $p \rightarrow q$ | $\neg p$ | $\neg p \vee q$ |
|---|---|---|---|---|
| T | T | T | F | T |
| T | F | F | F | F |
| F | T | T | T | T |
| F | F | T | T | T |

∴ $p \rightarrow q \equiv \neg p \vee q$

29) $a * b = a + b + ab - 7$.

Sum and difference of real number is a real number.

$a + b + ab - 7$ is a real number.

∴ $a * b \in R$.

∴ $*$ is a binary on $R$

30) $I = \int_0^{\pi/2} \frac{\sin x \, dx}{1 + \cos^2 x}$.

Put $u = \cos x \Rightarrow du = -\sin x \, dx$

$-du = \sin x \, dx$.

When $x = 0$ $u = \cos 0 = 1$

$x = \pi/2$ $u = \cos \pi/2 = 0$

$I = \int_1^0 \frac{-du}{1+u^2} = \int_0^1 \frac{du}{1+u^2}$

$= [\tan^{-1}(u)]_0^1$

$= \tan^{-1}(1) - \tan^{-1}(0)$

$= \pi/4 - 0 = \frac{\pi}{4}$.

31) Let $w = e^{i\frac{2\pi}{9}}$.

∴ $1, w, w^2 \ldots w^8$ are 9th root of unity.

∴ $1 + w + w^2 + \ldots + w^8 = 0$

$\therefore w + w^2 + \ldots + w^8 = -1$

$\Rightarrow \sum_{k=1}^{8} e^{i\frac{2k\pi}{9}} = -1$

32) The sign of coefficient of $f(x)$ is

$+ - + + +$

The number of changes in sign is 2.

∴ maximum number of positive roots is 2.

The sign of coefficient of $f(-x)$ is

$- + + + +$

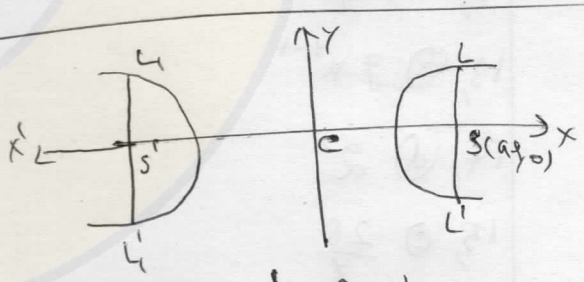
number of change in sign is 1.

∴ number of real root is one.

∴ maximum number of real root is 3.

∴ minimum number of imaginary roots are 6.

33)




foci are $S(ae, 0)$ $S'(-ae, 0)$

Coord of $L = L(ae, y_1)$.

$\therefore \frac{a^2e^2}{a^2} - \frac{y_1^2}{b^2} = 1$

$e^2 - 1 = \frac{y_1^2}{b^2}$

$\therefore y_1^2 = b^2(e^2 - 1)$

$= b^2 \times \frac{b^2}{a^2}$

$y_1 = \frac{b^2}{a}$

∴ Length of $LR = LL' = 2y_1 = \frac{2b^2}{a}$

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34) Volume $= V = [\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} -2 & 5 & 3 \\ 1 & 3 & -2 \\ -3 & 1 & 4 \end{vmatrix}$

$= 12$

$\vec{b} \times \vec{c} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & -2 \\ -3 & 1 & 4 \end{vmatrix} = 14\vec{i} + 2\vec{j} + 10\vec{k}$

$|\vec{b} \times \vec{c}| = \sqrt{196 + 4 + 100} = \sqrt{300} = 10\sqrt{3}$

length of altitude $= h = \frac{\text{volume}}{\text{area}}$

$= \frac{12}{10\sqrt{3}} = \frac{4 \times 3}{10 \times \sqrt{3}}$

$= \frac{2\sqrt{3}}{5}$

35) $f(x) = x^4 - 4x^3$

$f'(x) = 4x^3 - 12x^2$

$f''(x) = 12x^2 - 24x$

$= 12x(x-2)$

Put $f''(x) = 0 \Rightarrow x = 0, 2$

intervals are $(-\infty, 0)\ (0, 2)\ (2, \infty)$

when $x < 0$ $f''(x) > 0$

$f$ is concave up in $(-\infty, 0)$

when $0 < x < 2$ $f''(x) < 0$

$f$ is concave down in $(0, 2)$.

when $x > 2$ $f''(x) > 0$

$f$ is concave up in $(2, \infty)$

$\therefore$ $f$ is concave up in $(-\infty, 0) \cup (2, \infty)$ and concave down in $(0, 2)$.

36) $f(x) = Ax^2 + Bx + C$

$f'(x) = 2Ax + B$.

$f'(x) = \frac{f(b) - f(a)}{b - a}$

$2Ax + B = \frac{(Ab^2 + Bb + C) - (Aa^2 + Ba + C)}{b - a}$

$= \frac{A(b^2 - a^2) + B(b - a)}{b - a}$

$= A(b + a) + B$

$\therefore$ $2x = b + a \Rightarrow x = \frac{a+b}{2}$.

37) $I = \int_0^{\pi/2} \sin^2 x\,dx + \int_0^{\pi/2} \cos^4 x\,dx$.

$= \frac{1}{2} \times \frac{\pi}{2} + \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$

$= \frac{\pi}{4}\left(1 + \frac{3}{4}\right) = \frac{\pi}{4} \times \frac{7}{4} = \frac{7\pi}{16}$

38) $V = \frac{4\pi}{3} r^3$ ; $S = 4\pi r^2$

Given $\frac{dV}{dt} \propto S$

$\therefore$ $\frac{dV}{dt} = -kS$

$\frac{4\pi}{3}\left(3r^2 \frac{dr}{dt}\right) = -k(4\pi r^2)$

$4\pi r^2 \frac{dr}{dt} = -k(4\pi r^2)$

$\frac{dr}{dt} = -k$ is the D.E

39) $0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$10k^2 + 9k = 1$

$10k^2 + 9k - 1 = 0$

$(10k - 1)(k + 1) = 0$

$k = \frac{1}{10}, -1$.

$\therefore$ $k = \frac{1}{10}$

$P(0 < x < 4) = k + 2k + 2k$

$= 5k = \frac{1}{2}$.

40) $A = \begin{pmatrix} 3 & -2 \\ \lambda & -2 \end{pmatrix} \Rightarrow A^2 = \begin{pmatrix} 3 & -2 \\ \lambda & -2 \end{pmatrix}\begin{pmatrix} 3 & -2 \\ \lambda & -2 \end{pmatrix}$

$= \begin{pmatrix} 9 - 2\lambda & -2 \\ \lambda & -2\lambda + 4 \end{pmatrix}$

$\lambda A - 2I = \begin{pmatrix} 3\lambda & -2\lambda \\ \lambda^2 & -2\lambda \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

$= \begin{pmatrix} 3\lambda - 2 & -2\lambda \\ \lambda^2 & -2\lambda - 2 \end{pmatrix}$

$\therefore$ $\begin{pmatrix} 9 - 2\lambda & -2 \\ \lambda & -2\lambda + 4 \end{pmatrix} = \begin{pmatrix} 3\lambda - 2 & -2\lambda \\ \lambda^2 & -2\lambda - 2 \end{pmatrix}$

$\therefore$ $-2 = -2\lambda \Rightarrow \lambda = 1$

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Part - IV

41 (a) $\begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & \lambda \\ 1 & 3 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7 \\ \mu \\ 5 \end{pmatrix}$

$\therefore [A/B] = \begin{bmatrix} 1 & 2 & 1 & 7 \\ 1 & 1 & \lambda & \mu \\ 1 & 3 & -5 & 5 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 2 & 1 & 7 \\ 0 & -1 & \lambda-1 & \mu-7 \\ 0 & 1 & -6 & -2 \end{bmatrix} \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix}$

$\sim \begin{bmatrix} 1 & 2 & 1 & 7 \\ 0 & 1 & -6 & -2 \\ 0 & -1 & \lambda-1 & \mu-7 \end{bmatrix} R_2 \leftrightarrow R_3$

$\sim \begin{bmatrix} 1 & 2 & 1 & 7 \\ 0 & 1 & -6 & -2 \\ 0 & 0 & \lambda-7 & \mu-9 \end{bmatrix} R_3 \to R_2 + R_3$

(i) If $\lambda = 7$, $\mu \neq 9$ then $\rho(A) = 2$

$\rho[A/B] = 3 \quad \therefore \rho(A) \neq \rho[A/B]$

The equations are inconsistent and has no solution.

(ii) If $\lambda \neq 7$, $\mu$ is any real number.

$\therefore \rho(A) = \rho[A/B] = 3$

The equations are consistent and has unique solution.

(OR)

41 (b) Let $\alpha, \beta, \gamma$ and $\delta$ are the roots of given equations.

$\therefore \alpha+\beta+\gamma+\delta = -\frac{b}{a} = \frac{8}{2} = 4.$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a} = \frac{6}{2} = 3$

$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -\frac{d}{a} = \frac{0}{2}$

$= 0$

$\alpha\beta\gamma\delta = \frac{e}{a} = -\frac{3}{2}$

$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha+\beta+\gamma+\delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$

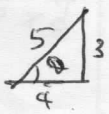
$= (4)^2 - 2(3)$

$= 16 - 6 = 10$

42 (a) $\sin\left[\sin^{-1}\left(\frac{3}{5}\right) + \sec^{-1}\left(\frac{5}{4}\right)\right]$

$= \sin\left[\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{4}{5}\right)\right]$

Let $\theta = \cos^{-1}\left(\frac{4}{5}\right) \Rightarrow \cos\theta = \frac{4}{5}$




$\therefore \sin\theta = \frac{3}{5}$

$\theta = \sin^{-1}\left(\frac{3}{5}\right)$

L.H.S

$= \sin(\theta + \theta)$

$= \sin(2\theta)$

$= 2\sin\theta\cos\theta$

$= 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$

(OR)

42 (b) $4x^2 - 25y^2 - 24x + 250y - 489 = 0$

$4(x^2 - 6x) - 25(y^2 - 10y) = 489.$

$4[(x-3)^2 - 9] - 25[(y-5)^2 - 25] = 489$

$4(x-3)^2 - 36 - 25(y-5)^2 + 625 = 489$

$4(x-3)^2 - 25(y-5)^2 = 489 + 36 - 625$

$= -100$

$\frac{(y-5)^2}{4} - \frac{(x-3)^2}{25} = 1$

$h = 3, \quad k = 5$

$a^2 = 4 \quad b^2 = 25 \Rightarrow a = 2, \ b = 5$

$e^2 = \frac{a^2+b^2}{a^2} = \frac{29}{4} \Rightarrow e = \frac{\sqrt{29}}{2}$

$ae = 2 \times \frac{\sqrt{29}}{2} = \sqrt{29}.$

(i) foci $= F(h, ae+k); F'(h, -ae+k)$

$= F(3, \sqrt{29}+5); F'(3, -\sqrt{29}+5)$

(ii) Vertices $= A(h, a+k); A'(h, -a+k)$

$= A(3, 2+5); A'(3, -2+5)$

$= A(3, 7) \quad A'(3, 3)$

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42) (a) $\vec{a} = \vec{i} - 2\vec{j} + 4\vec{k}$; $\vec{u} = \vec{i} + 2\vec{j} - 3\vec{k}$;
$\vec{v} = 3\vec{i} - \vec{j} + \vec{k}$.

V.E $\vec{r} = \vec{a} + s\vec{u} + t\vec{v}$
$= (\vec{i} - 2\vec{j} + 4\vec{k}) + s(\vec{i} + 2\vec{j} - 3\vec{k})$
$+ t(3\vec{i} - \vec{j} + \vec{k})$.

C.E $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$

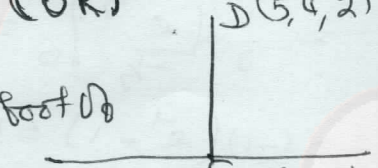
$(x_1, y_1, z_1) = (1, -2, 4)$ $(b_1\ b_2\ b_3) = (1, 2, -3)$
$(c_1\ c_2\ c_3) = (3, -1, 1)$.

$\begin{vmatrix} x-1 & y+2 & z-4 \\ 1 & 2 & -3 \\ 3 & -1 & 1 \end{vmatrix} = 0$

$x + 10y + 7z - 9 = 0$

(OR)

43) (b)




Let F is the foot of ⊥r of the line DF.

The coordinate of $F = F(2t-1, 3t+3, -t+1)$.

$\vec{DF} = \vec{OF} - \vec{OD} = (2t-6)\vec{i} + (3t-1)\vec{j} + (-t-1)\vec{k}$.

$\because \vec{b} \perp r\ \vec{DF}$.

$2(2t-6) + 3(3t-1) - (-t-1) = 0$

$t = 1$.

∴ Coordinate of $f = f(1, 6, 0)$

Equation of ⊥r DF is

$\frac{x-5}{-4} = \frac{y-4}{2} = \frac{z-2}{-2}$

44) (a) $f(x) = \frac{1}{2}(e^x - e^{-x})$

$f'(x) = \frac{1}{2}(e^x + e^{-x})$

$f''(x) = \frac{1}{2}(e^x - e^{-x})$.

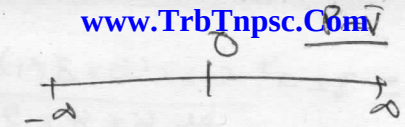
Put $f''(x) = 0 \Rightarrow \frac{1}{2}(e^x - e^{-x}) = 0$

$e^x = e^{-x}$

$x = -x \Rightarrow 2x = 0$

$x = 0$

Intervals are $(-\infty, 0)$ $(0, \infty)$




when $x < 0$ (say $x = -1$)

$f''(-1) = \frac{1}{2}(e^{-1} - e)$

$= \frac{1}{2}\left(\frac{1-e^2}{e}\right) < 0$

∴ gf is concave down in $(-\infty, 0)$

when $x > 0$ (say $x = 1$)

$f''(1) = \frac{1}{2}(e - e^{-1}) = \frac{1}{2}\left(\frac{e^2-1}{e}\right) > 0$.

∴ gf is concave up in $(0, \infty)$.

At $x = 0$, $f''(x)$ change its sign

∴ $x = 0$ is a point of inflection.

$y = \frac{1}{2}(1-1) = 0$

∴ Point of inflection $= (0, 0)$

(OR)

44) (b) $\int_0^{2a} x^2\sqrt{2ax - x^2}\,dx$.

Put $x = 2a\cos^2\theta \Rightarrow dx = 2a \times 2(\cos\theta)(-\sin\theta)\,d\theta$

$= -4a\sin\theta\cos\theta\,d\theta$

when $x = 0$ $\theta = \frac{\pi}{2}$

$x = 2a$ $\theta = 0$

∴ $I = \int_{\pi/2}^{0} 4a^2\cos^4\theta\sqrt{4a^2\cos^2\theta - 4a^2\cos^4\theta}\,(-4a\cos\theta\sin\theta)\,d\theta$

$= \int_{\pi/2}^{0} 4a^2\cos^4\theta\,(2a\cos\theta\sin\theta)\,(-4a\cos\theta\sin\theta)\,d\theta$

$= 32a^4\int_0^{\pi/2}\sin^2\theta\cos^6\theta\,d\theta$

By Bernulli's formula

$\int_0^{\pi/2}\sin^m\theta\cos^n\theta\,d\theta = \frac{(m-1)(m-3)\cdots 2\text{ or }1}{(m+n)(m+n-2)\cdots 2\text{ or }1} \times \frac{\pi}{2}$

$= \frac{(2-1)\times(5\times 3\times 1)}{8\times 6\times 4\times 2} \times \frac{\pi}{2}$

$= 32a^4 \times \frac{(2-1)\times(5\times 3\times 1)}{8\times 6\times 4\times 2} \times \frac{\pi}{2}$

(m, n are even).

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$= 32a^4 \frac{(1)\times(5\times3\times1)}{8\times6\times4\times2} \times \frac{\pi}{2}$

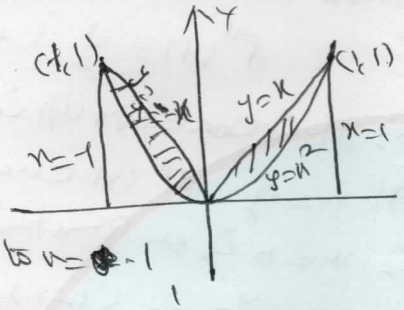
$= \frac{5\pi a^4}{8}$.

---

45) (a) $x^2 = y$

$y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x \leq 0 \end{cases}$.

Point of intersection are $(1,1)$ $(-1,1)$.




limit $x=-1$ to $x=1$

Required Area $= A = 2\int_0^1 (y_1 - y_2)\,dx$

$= 2\int_0^1 (x - x^2)\,dx$

$= 2\left(\frac{x^2}{2} - \frac{x^3}{3}\right)_0^1 = 2\left(\frac{1}{2} - \frac{1}{3} - 0\right)$

$= 2\left(\frac{1}{6}\right) = \frac{1}{3}$ sq units

(OR)

45) (b) $L\frac{di}{dt} + Ri = E$

$\frac{di}{dt} + \left(\frac{R}{L}\right)i = \frac{E}{L}$.

Here $P = \frac{R}{L}$; $Q = \frac{E}{L}$.

$\int P\,dt = \int \frac{R}{L}\,dt = \frac{R}{L}t$.

$\therefore$ I.F $= e^{\int P\,dt} = e^{\frac{R}{L}t}$.

Solution is $i\,e^{\frac{R}{L}t} = \int \frac{E}{L} e^{\frac{R}{L}t}\,dt + c$

$= \frac{E}{L} \times \frac{e^{\frac{R}{L}t}}{\frac{R}{L}} + c$

$= \frac{E}{R} e^{\frac{R}{L}t} + c$

$\therefore i = \frac{E}{R} + c\,e^{-\frac{R}{L}t}$.

When $E = 0$, $i = c\,e^{-\frac{R}{L}t}$

---

46) (a) $\because f(x)$ is a P.d.f

(i) $\int_0^\infty f(x)\,dx = 1$

$\int_0^\infty k\,e^{-x/3}\,dx = 1$

$k\left[\frac{e^{-x/3}}{-1/3}\right]_0^\infty = 1$

$k\left[-3e^{-x/3}\right]_0^\infty = 1$

$-3k[0-1] = 1$

$3k = 1 \Rightarrow k = \frac{1}{3}$ $\therefore f(x) = \frac{1}{3}e^{-x/3}$

(ii) $P(x<3) = \int_0^3 f(x)\,dx$

$= \int_0^3 \frac{1}{3} e^{-x/3}\,dx$

$= \frac{1}{3}\left[\frac{e^{-x/3}}{-1/3}\right]_0^3 = (-1)\left[e^{-x/3}\right]_0^3$

$= (-1)\left(e^{-1} - 1\right) = 1 - e^{-1}$.

(OR)

46) (b)

| $p$ | $q$ | $p \to q$ | $\sim p$ | $\exists \to \sim p$ | $\exists \to \sim p) \Theta (\exists \to q)$ |
|---|---|---|---|---|---|
| T | T | T | F | T | T |
| T | F | F | F | T | F |
| F | T | T | T | T | T |
| F | F | T | T | F | F. |

The last column contains not all T (or) all F. $\therefore$ It is a contingency.

---

47) (a) $z^4 = (1 - i\sqrt{3}) \Rightarrow z = (1 - i\sqrt{3})^{1/4}$

$1 - i\sqrt{3} = r(\cos\theta + i\sin\theta)$

$r\cos\theta = 1$

$r\sin\theta = -\sqrt{3}$

$r^2 = 1 + 3 = 4 \Rightarrow r = 2$

$\left.\begin{aligned} \cos\theta &= \frac{1}{2} \\ \sin\theta &= \frac{-\sqrt{3}}{2} \end{aligned}\right\} \Rightarrow \theta = -\frac{\pi}{3}$ ($\because \theta$ lies in 4th quadrant)

$1 - i\sqrt{3} = 2\left[\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right]$

$= 2\left[\cos\left(2k\pi - \frac{\pi}{3}\right) + i\sin\left(2k\pi - \frac{\pi}{3}\right)\right]$

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$(1-i\sqrt{3})^{1/4} = 2^{1/4}\left[\cos\left(2k\pi-\frac{\pi}{3}\right)+i\sin\left(2k\pi-\frac{\pi}{3}\right)\right]^{1/4}$

$= 2^{1/4}\left[\cos\frac{1}{4}\left(\frac{6k\pi-\pi}{3}\right)+i\sin\frac{1}{4}\left(\frac{6k\pi-\pi}{3}\right)\right]$; $k = 0, 1, 2, 3$.

When $k=0$ $\quad 2^{1/4}\,\mathrm{cis}\left(\frac{-\pi}{12}\right)$

$k=1$ $\quad 2^{1/4}\,\mathrm{cis}\,\frac{5\pi}{12}$

$k=2$ $\quad 2^{1/4}\,\mathrm{cis}\,\frac{11\pi}{12}$

$k=3$ $\quad 2^{1/4}\,\mathrm{cis}\,\frac{17\pi}{12}$

---

47, (6) $f = \log\sqrt{x^2+y^2} = \frac{1}{2}\log(x^2+y^2)$ (OR)

$f_x = \frac{1}{2}\times\frac{1}{x^2+y^2}(2x) = \frac{x}{x^2+y^2}$

$f_{xx} = \frac{(x^2+y^2)(1)-x(2x)}{(x^2+y^2)^2}$

$= \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2} \to (1)$

$f_y = \frac{1}{2}\frac{1}{x^2+y^2}(2y) = \frac{y}{x^2+y^2}$

$f_{yy} = \frac{(x^2+y^2)(1)-y(2y)}{(x^2+y^2)^2}$

$= \frac{x^2+y^2-2y^2}{(x^2+y^2)^2}$

$= \frac{x^2-y^2}{(x^2+y^2)^2} = -\frac{(y^2-x^2)}{(x^2+y^2)^2} \to (2)$

(1) + (2) $\quad f_{xx}+f_{yy} = \frac{y^2-x^2}{(x^2+y^2)^2} - \frac{(y^2-x^2)}{(x^2+y^2)^2}$

$= 0$

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## PTA Model Question Paper – 3 Answer key (EM) 2019 - 2020

### Part – I

| Answer all the questions | |
|---|---|
| 1. | b) $(\cos^2\frac{\theta}{2})$ |
| 2. | d) $e^{(\Delta_1/\Delta_3)}, e^{(\Delta_2/\Delta_3)}$ |
| 3. | b) imaginary axis |
| 4. | d) $\frac{\pi}{2}$ |
| 5. | a) One negative and 2 imaginary zeros |
| 6. | d) $\tan^{-1}\left(\frac{1}{2}\right)$ |
| 7. | $\left(0,\frac{1}{8}\right)$ |
| 8. | a) $2ab$ |
| 9. | a) 0 |
| 10. | d) $3, -9$ |
| 11. | c) 12 |
| 12. | c) 3 |
| 13. | a) $\frac{2}{(x+1)^2}dx$ |
| 14. | a) $\frac{1}{2}$ |
| 15. | a) 0 |
| 16. | c) 1 |
| 17. | a) $P = ce^{kt}$ |
| 18. | b) $\left(25,\frac{1}{5}\right)$ |
| 19. | d) $\frac{2}{3}$ |
| 20. | d) $\neg(p \wedge q) \rightarrow \neg(p \vee q)$ |

### Part – II

21. TB : 64 (Example: 2.8 - (i))
22. TB : 137 (Ex. 4.1 – 4 (ii))
23. TB : 182 (Ex. 5.1 - 5)
24. TB : 259 (Ex. 6.6 - 5)
25. CQ
26. TB : 120 (Ex. 9.6 – 1 (i))
27. TB : 157 (Ex. 10.4 - 4)
28. TB : 210 (Ex. 11.4 – 6)
29. TB : 235 (Ex. 12.1 – 2)
30. CQ

### Part - III

31. TB : 33 (Ex. 1.3 – 1 (ii))
32. CQ
33. CQ
34. TB : 276 (Ex. 6.9 – 1)
35. TB : 337 (Example: 7.52)
36. TB : 79 (Ex. 8.4 – 4)
37. TB : 114 (Example: 9.31)
38. CQ
39. TB : 216 (Example : 11.21 (i))
40. CQ

## PTA Model Question Paper – 3 Answer key (TM) 2019 - 2020

### பகுதி – I

| அனைத்து வினாக்களுக்கும் விடையளி | |
|---|---|
| 1. | b) $(\cos^2\frac{\theta}{2})$ |
| 2. | d) $e^{(\Delta_1/\Delta_3)}, e^{(\Delta_2/\Delta_3)}$ |
| 3. | b) கற்பனை அச்சு |
| 4. | d) $\frac{\pi}{2}$ |
| 5. | a) ஒரு குறை மற்றும் இரு மெய்யற்ற கலப்பெண் பூஜ்ஜியமாக்கிகள் இருக்கும். |
| 6. | d) $\tan^{-1}\left(\frac{1}{2}\right)$ |
| 7. | $\left(0,\frac{1}{8}\right)$ |
| 8. | a) $2ab$ |
| 9. | a) 0 |
| 10. | d) $3, -9$ |
| 11. | c) 12 |
| 12. | c) 3 |
| 13. | a) $\frac{2}{(x+1)^2}dx$ |
| 14. | a) $\frac{1}{2}$ |
| 15. | a) 0 |
| 16. | c) 1 |
| 17. | a) $P = ce^{kt}$ |
| 18. | b) $\left(25,\frac{1}{5}\right)$ |
| 19. | d) $\frac{2}{3}$ |
| 20. | d) $\neg(p \wedge q) \rightarrow \neg(p \vee q)$ |

### பகுதி – II

21. TB : 70 (எகா. 2.8 - (i))
22. TB : 151 (பயிற்சி 4.1 – 4 (ii))
23. TB : 200 (பயிற்சி 5.1 - 5)
24. TB : 284 (பயிற்சி 6.6 - 5)
25. CQ
26. TB : 125 (பயிற்சி 9.6 – 1 (i))
27. TB : 166 (பயிற்சி 10.4 - 4)
28. TB : 222 (பயிற்சி 11.4 – 6)
29. TB : 250 (பயிற்சி 12.1 – 2)
30. CQ

### பகுதி – III

31. TB : 36 (பயிற்சி 1.3 – 1 (ii))
32. CQ
33. CQ
34. TB : 303 (பயிற்சி 6.9 – 1)
35. TB : 38 (எகா. 7.52)
36. TB : 83 (பயிற்சி 8.4 – 4)
37. TB : 119 (எகா. 9.31)
38. CQ
39. TB : 228 (எகா. 11.21 (i))



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| Part – IV | பகுதி – IV |
|---|---|
| 41. a) TB : 37 (Ex. 1.5 – 2)<br>b) TB : 85 (Example: 2.31)<br>42. a) TB : 122 (Example: 3.28)<br>b) CQ<br>43. a) TB : 194 (Example: 5.21)<br>b) TB : 226 (Example: 6.3)<br>44. a) TB : 40 (Ex. 7.6 – 1 (ii))<br>b) TB : 79 (Ex. 8.4 – 2 (i))<br>45. a) TB : 132 (Example: 9.60)<br>b) CQ<br>46. a) TB : 194(Ex. 11.2 - 5)<br>b) TB : 249 (Ex. 12.2 - 13)<br>47. a) CQ<br>b) TB : 46 (Example: 7.65) | 40. CQ<br>41. a) TB : 41 (பயிற்சி 1.5 – 2)<br>b) TB : 91 (எகா. 2.31)<br>42. a) TB : 133 (எகா. 3.28)<br>b) CQ<br>43. a) TB : 211 (எகா. 5.21)<br>b) TB : 246 (எகா. 6.3)<br>44. a) TB : 41 (பயிற்சி 7.6 – 1 (ii))<br>b) TB : 83 (பயிற்சி 8.4 – 2 (i))<br>45. a) TB : 138 (எகா. 9.60)<br>b) CQ<br>46. a) TB : 206 (பயிற்சி 11.2 - 5)<br>b) TB : 265 (பயிற்சி 12.2 - 13)<br>47. a) CQ<br>b) TB : 48 (எகா. 7.65) |

CQ – Creative Question



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12 Std Maths P.T.A Modal question Paper - 3 [P - I]

Answer

Part - I

1. (b) $\left(\cos^2\frac{\theta}{2}\right)A^+$.
2. (d) $e^{\left(\frac{D_1}{D_3}\right)}, e^{\left(\frac{D_2}{D_3}\right)}$
3. (b) Imaginary axis.
4. (d) $\frac{\pi}{2}$
5. (a) one negative and two real roots.
6. (d) $\tan^{-1}\left(\frac{1}{2}\right)$
7. Vertex of Parabola $x^2 = 8y$ is $(0, \frac{1}{8})$
8. (a) $2ab$.
9. (a) $\lambda + \mu = 0$
10. (d) $3; -9$
11. (c) 12
12. (c) 3
13. (a) $\frac{2}{(x+1)^2}dx$.
14. (a) $\frac{1}{2}$
15. (a) 0
16. (c) 1
17. (a)
18. (b) $(25; \frac{1}{5}) = (h, k)$
19. (d) $\frac{2}{3}$
20. (d) $(\neg p \wedge \neg q) \rightarrow (\neg p \vee q)$

Part - II

21. $\overline{z} = \overline{(2+i\sqrt{3})^{60}} + \overline{(2-i\sqrt{3})^{60}}$
$= \overline{(2+i\sqrt{3})}^{60} + \overline{(2-i\sqrt{3})}^{60}$
$= (2-i\sqrt{3})^{60} + (2+i\sqrt{3})^{60} = z$
$\therefore z = \overline{z}$ $\therefore z$ is real

22. $\sin^{-1}\left(\sin\left(\frac{5\pi}{4}\right)\right) = \sin^{-1}\left(\sin\left(\pi + \frac{\pi}{4}\right)\right)$
$= \sin^{-1}\left(-\sin\frac{\pi}{4}\right)$
$= \sin^{-1}\left(\sin\left(-\frac{\pi}{4}\right)\right)$
$= -\frac{\pi}{4}$

23. Equation of the circle with (3, 4) and (2, -7) as end of diameter is
$(x-3)(x-2) + (y-4)(y+7) = 0$
$x^2 - 5x + 6 + y^2 + 3y - 28 = 0$
$x^2 + y^2 - 5x + 3y - 22 = 0$

24. Equation of the plane $6x + 4y - 3z = 12$
$\frac{x}{2} + \frac{y}{3} - \frac{z}{4} = 1$
i.e. $\frac{x}{a} + \frac{y}{b} + \frac{z}{-4} = 1$
$\therefore a = 2; b = 3; c = -4$

25. $f(x) = x^4 - 2x^2$
$f'(x) = 4x^3 - 4x$
$f'(c) = 4c^3 - 4c = 4c(c^2 - 1)$
$f'(c) = 0 \Rightarrow 4c(c^2-1) = 0$
$4c = 0 ; c^2 - 1 = 0$
$c = 0 ; c^2 = 1$
$c = \pm 1$
$\therefore c = -1, 0, 1$

26. $\int_0^{\pi/2} \sin^{10} x\, dx = \frac{9}{10} \times \frac{7}{8} \times \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$
$= \frac{63 \times 5\pi}{32 \times 16}$
$= \frac{63\pi}{32 \times 16} = \frac{63\pi}{512}$

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27) $y = e^{-x} + mx + n$

$\frac{dy}{dx} = -e^{-x} + m$

$\frac{d^2y}{dx^2} = -(-e^{-x}) + 0 = e^{-x}$

$\therefore e^x\left(\frac{d^2y}{dx^2}\right) - 1 = e^x e^{-x} - 1 = 1 - 1 = 0$

---

28) $\text{mean} = E(x) = \int_0^\infty x f(x)\,dx$

$= \int_0^\infty x(3e^{-3x})\,dx$

$= 3\int_0^\infty x e^{-3x}\,dx$

By Gamma

$= 3\,\frac{\lfloor 1}{3^2} = 3 \times \frac{1}{3^2} = \frac{1}{3}$

---

29) $m \otimes n = m^n + n^m$

Put $m = 2 \quad n = -3$

$2^{-3} + (-3)^2 = \frac{1}{8} + 9 = \frac{73}{8} \notin Z$

$\therefore m^n + n^m \notin Z \Rightarrow m \otimes n \notin Z.$

$\otimes$ is not a Binary operation on Z.

---

30) $AB = \begin{pmatrix} 14 & 15 \\ 8 & 10 \end{pmatrix} \Rightarrow \text{Adj}(AB) = \begin{pmatrix} 10 & -15 \\ -8 & 14 \end{pmatrix}$

---

## Part - III

31) $\begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \end{pmatrix}$

$AX = B. \Rightarrow X = A^{-1}B$

$A = \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix} \quad X = \begin{pmatrix} x \\ y \end{pmatrix} \quad B = \begin{pmatrix} 8 \\ -2 \end{pmatrix}$

$|A| = 7 \qquad \text{Adj}A = \begin{pmatrix} 2 & 1 \\ -3 & 2 \end{pmatrix}$

$A^{-1} = \frac{1}{7}\begin{pmatrix} 2 & 1 \\ -3 & 2 \end{pmatrix}$

$X = A^{-1}B = \frac{1}{7}\begin{pmatrix} 2 & 1 \\ -3 & 2 \end{pmatrix}\begin{pmatrix} 8 \\ -2 \end{pmatrix}$

$= \frac{1}{7}\begin{pmatrix} 14 \\ -28 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$

$x = 2; \quad y = -4$

P - II

---

32) $L.H.S = \frac{\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}}{\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}}$

$= \frac{\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}}{\cos(-\pi/3) + i\sin(-\pi/3)}$

$= \cos\left(\frac{\pi}{6} + \frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{6} + \frac{\pi}{3}\right)$

$= \cos\frac{\pi}{2} + i\sin\pi/2$

$= 0 + i(1) = i$

---

33) foci $= F(3, 5); F'(-3, 5)$

centre $= C(0, 5) = (h, k)$

$e = 2$

$FF' = 2ae = \sqrt{6^2 + 0}$

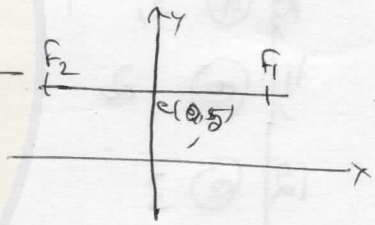
$= \sqrt{36}$

$2ae = 6$

$ae = 3$

$a(2) = 3 \Rightarrow a = 3/2$

$a^2 = \frac{9}{4}.$




$b^2 = (ae)^2 - a^2 = 9 - \frac{9}{4} = \frac{27}{4}.$

Equation of the hyperbola

$\frac{(x-0)^2}{\frac{9}{4}} - \frac{(y-5)^2}{\frac{27}{4}} = 1$

$\frac{x^2}{\frac{9}{4}} - \frac{(y-5)^2}{\frac{27}{4}} = 1$

$\frac{4x^2}{9} - \frac{4(y-5)^2}{27} = 1$

---

34) Equation of given planes

$2x - 7y + 4z - 3 = 0; \quad 3x - 5y + 4z + 11 = 0$

Equation of required plane

$(2x - 7y + 4z - 3) + \lambda(3x - 5y + 4z + 11) = 0$

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∵ the plane passes through (−2, 1, 3).

$(-4-7+12-3)+\lambda(-6-5+12+11)=0$

$(-2)+\lambda(12)=0$

$-1+6\lambda=0 \Rightarrow \lambda=\frac{1}{6}$.

Equation of the plane

$(2x-7y+4z-3)+\frac{1}{6}(3x-5y+4z+11)=0$

$12x-42y+24z-18+3x-5y+4z+11=0$

$15x-47y+28z-7=0$.

---

35) $f(x)=x-\sin x$

$f'(x)=1-\cos x \geq 0$

∵ $f(x)$ is increasing

There is no sign change in $f'(x)$.

∴ It has no maximum (or) minimum point.

---

36) $u=\log(x^3+y^3+z^3)$

$\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=\frac{3x^2}{x^3+y^3+z^3}+\frac{3y^2}{x^3+y^3+z^3}+\frac{3z^2}{x^3+y^3+z^3}$

$=\frac{3(x^2+y^2+z^2)}{x^3+y^3+z^3}$

---

37) $I=\int_0^{\pi} x^2\cos nx\,dx$

$u=x^2 \quad dv=\cos nx\,dx$

$u=\frac{\sin nx}{n}$

By Bernoulli

$I=\left(x^2\frac{\sin nx}{n}-(2x)\frac{1}{n}\left(-\frac{\cos nx}{n}\right)+(2)\frac{1}{n^2}\left(\frac{\sin nx}{n}\right)\right|_0^{\pi}$

$=\left(0+\frac{2\pi}{n^2}(\cos\pi)^n+0\right)-(0)$

$=\frac{2\pi}{n^2}(-1)^n$. ∵ $\sin n\pi=0$, $\cos n\pi=-1$

---

38) $\frac{dy}{dx}=e^{x+y}+x^2e^{x^3+y}$

$=e^xe^y+x^2e^{x^3}e^y$

Divide by $e^y$

$e^{-y}dy=(e^x+x^2e^{x^3})dx$

$=e^xdx+e^{x^3}(x^2dx)$

$e^{-y}dy=e^xdx+\frac{1}{3}e^{x^3}d(x^3)$

$\int e^{-y}dy=\int e^xdx+\frac{1}{3}\int e^{x^3}d(x^3)$

$-e^{-y}=e^x+\frac{1}{3}e^{x^3}+c$

---

39) mean $=np=2$

variance $=npq=1.5$

$q=\frac{npq}{np}=\frac{1.5}{2}=\frac{15}{20}=\frac{3}{4}$

$p=1-q=1-\frac{3}{4}=\frac{1}{4}$.

$np=2$

$n(\frac{1}{4})=2 \Rightarrow n=8$.

$P(x=x)={}^8C_x\left(\frac{1}{4}\right)^x\left(\frac{3}{4}\right)^{8-x}$; $x=0,1,\ldots,8$

$P(x=0)={}^8C_0\left(\frac{1}{4}\right)^0\left(\frac{3}{4}\right)^8$

$=(1)(1)\left(\frac{3}{4}\right)^8$

$=\left(\frac{3}{4}\right)^8$

---

40) $\vec{f}=\vec{i}+\vec{j}+\vec{k}$.

P.V of through Point $=0\vec{i}+0\vec{j}+0\vec{k}$.

P.V of about Point $=0\vec{i}-2\vec{j}+3\vec{k}$.

$\vec{r}=0\vec{i}+2\vec{j}-3\vec{k}$.

moment $=\vec{r}\times\vec{f}$.

$\vec{r}\times\vec{f}=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\0&2&-3\\1&1&1\end{vmatrix}=\vec{i}(5)-\vec{j}(3)+\vec{k}(-2)$

$=5\vec{i}-3\vec{j}-2\vec{k}$.

magnitude $=\sqrt{25+9+4}=\sqrt{38}$

D.C $=\left(\frac{5}{\sqrt{38}};\frac{-3}{\sqrt{38}};\frac{-2}{\sqrt{38}}\right)$.

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4, (a) $f(x) = ax^2+bx+c$

$f(-3) = 9 \Rightarrow 9a-3b+c = 9 \rightarrow (1)$

$f(5) = 61 \Rightarrow 25a+5b+c = 61 \rightarrow (2)$

$f(1) = 9 \Rightarrow a+b+c = 9 \rightarrow (3)$

$$[A|B] = \begin{bmatrix} 9 & -3 & 1 & 9 \\ 25 & 5 & 1 & 61 \\ 1 & 1 & 1 & 9 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 9 \\ 25 & 5 & 1 & 61 \\ 9 & -3 & 1 & 9 \end{bmatrix} R_1 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 9 \\ 0 & -20 & -24 & -164 \\ 0 & -12 & -8 & -60 \end{bmatrix} \begin{matrix} R_2 \leftrightarrow R_2 - 25R_1 \\ R_3 \leftrightarrow R_3 - 9R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 9 \\ 0 & 5 & 6 & 41 \\ 0 & 3 & 2 & 15 \end{bmatrix} \begin{matrix} R_2 \to -\frac{1}{4}R_2 \\ R_3 \to -\frac{1}{4}R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 9 \\ 0 & 5 & 6 & 41 \\ 0 & 0 & -8 & -48 \end{bmatrix} R_3 \leftrightarrow 5R_3 - 3R_2$$

$\therefore -8c = -48 \Rightarrow \underline{c = 6}$

$5b+6c = 41$

$5b+36 = 41 \Rightarrow 5b = 5$

$\underline{b = 1}$

$a+b+c = 9$

$a+1+6 = 9 \Rightarrow a = 9-7$

$\underline{a = 2}$

(OR)

41, (b) $-\sqrt{3}+3i = r(\cos\theta + i\sin\theta)$

$r\cos\theta = -\sqrt{3}$

$r\sin\theta = 3$

$r^2 = 3+9 = 12 = 4\times 3$

$r = 2\sqrt{3}$

$\left.\begin{matrix} \cos\theta = -\frac{1}{2} \\ \sin\theta = \frac{\sqrt{3}}{2} \end{matrix}\right\} \Rightarrow \theta = \pi - \frac{\pi}{3}$

$= \frac{2\pi}{3}$

$\therefore -\sqrt{3}+i(3) = 2\sqrt{3}\left[\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right]$.

$(-\sqrt{3}+3i)^{31} = (2\sqrt{3})^{31}\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)^{31}$

$= (2\sqrt{3})^{31}\left[\cos\frac{62\pi}{3} + i\sin\frac{62\pi}{3}\right]$

$= (2\sqrt{3})^{31}\left(\cos(20\pi+\frac{2\pi}{3}) + i\sin(20\pi+\frac{2\pi}{3})\right]$

$= (2\sqrt{3})^{31}\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right]$

$= (2\sqrt{3})^{31}\left(\cos(\pi-\frac{\pi}{3}) + i\sin(\pi-\frac{\pi}{3})\right]$

$= (2\sqrt{3})^{31}\left(-\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right]$

$= (2\sqrt{3})^{31}\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$.

42, (a) $x^4-10x^3+26x^2-10x+1 = 0$

$\div x^2$

$x^2-10x+26-10\frac{1}{x}+\frac{1}{x^2} = 0$

$(x^2+\frac{1}{x^2})-10(x+\frac{1}{x})+26 = 0$

$((x+\frac{1}{x})^2-2)-10(x+\frac{1}{x})+26 = 0$

Put $y = x+\frac{1}{x}$

$(y^2-2)-10y+26 = 0$

$y^2-10y+24 = 0$

$(y-4)(y-6) = 0$

$y = 4, y = 6$

Case (1) $y = 4$

$x+\frac{1}{x} = 4$

$x^2+1 = 4x$

$x^2-4x+1 = 0 \Rightarrow \underline{x = 2\pm\sqrt{3}}$

Case (2) $y = 6$

$x+\frac{1}{x} = 6$

$x^2+1 = 6x$

$x^2-6x+1 = 0 \Rightarrow \underline{x = 3\pm 2\sqrt{2}}$

(OR)

42, (b) $\tan^{-1}(x) + 2\cot^{-1}(x) = \frac{2\pi}{3}$

$\tan^{-1}(x) + \cot^{-1}(x) + \cot^{-1}(x) = \frac{2\pi}{3}$

$\frac{\pi}{2} + \cot^{-1}(x) = \frac{2\pi}{3}$

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$\cot(x) = \frac{\pi}{6}$

$x = \cot\frac{\pi}{6} = \sqrt{3}$

43. (c) $4x^2 + y^2 + 24x - 2y + 21 = 0$

$4(x^2 + 6x) + (y^2 - 2y) = -21$

$4[(x+3)^2 - 9] + [(y-1)^2 - 1] = -21$

$4(x+3)^2 - 36 + (y-1)^2 - 1 = -21$

$4(x+3)^2 + (y-1)^2 = -21 + 37$

$= 16$

$\frac{(x+3)^2}{4} + \frac{(y-1)^2}{16} = 1$

∴ Major axis Parallel to y-axis

$a^2 = 16 \Rightarrow a = 4$

$b^2 = 4$.

$e^2 = \frac{a^2 - b^2}{a^2} = \frac{12}{16} = \frac{3}{4} \Rightarrow e = \frac{\sqrt{3}}{2}$

$ae = 2\sqrt{3}$.

∴ $h = -3, k = 1$

(1) Centre $= C(h, k) = C(-3, 1)$

(2) Vertices $= A(h, k+a), A'(h, k-a)$

$= A(-3, 5)$ $A'(-3, -3)$

(3) Foci $= F(h, k+ae); F'(h, k-ae)$

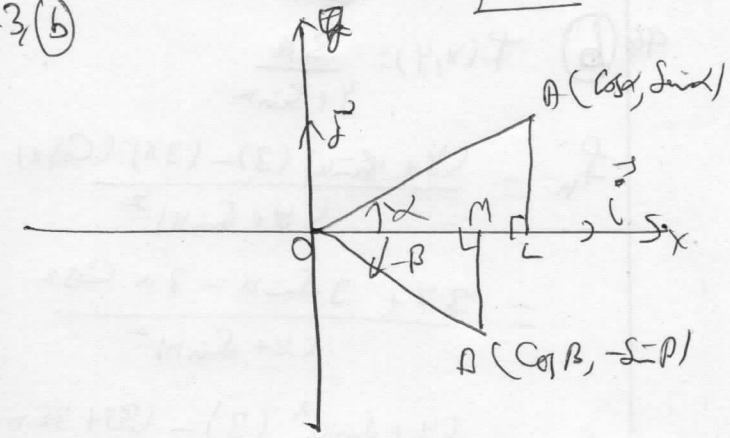
$= f(-3, 1+2\sqrt{3})$ $f'(-3, 1-2\sqrt{3})$

(4) Length of L.R $= \frac{2b^2}{a} = \frac{2(4)}{4}$

$= 2$

(OR)

43. (b)




$\vec{OA} = \cos\alpha\,\vec{i} + \sin\alpha\,\vec{j}$.

$\vec{OB} = \cos\beta\,\vec{i} - \sin\beta\,\vec{j}$.

$\vec{OA} \cdot \vec{OB} = (\cos\alpha\,\vec{i} + \sin\alpha\,\vec{j}) \cdot (\cos\beta\,\vec{i} - \sin\beta\,\vec{j})$

$= \cos\alpha\cos\beta - \sin\alpha\sin\beta \rightarrow (1)$

$\vec{OA} \cdot \vec{OB} = |\vec{OA}|\,|\vec{OB}| \cos(\alpha+\beta)$

$= 1 \times 1 \times \cos(\alpha+\beta)$

$= \cos(\alpha+\beta) \rightarrow (2)$

From (1), (2)

$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

44. (a) $a = -1, \; b = 2$

$f(x) = 3x^4 - 4x^3$

$f'(x) = 12x^3 - 12x^2$

Put $f'(c) = 0$

$12c^3 - 12c^2 = 0$

$12c^2(c-1) = 0$

$c = 0, 1$

i.e. $c = 0, 1$

$f(a) = f(-1) = 3(1) - 4(-1)$

$= 3 + 4 = 7$

$f(c) = f(0) = 0$

$f(c) = f(1) = 3 - 4 = -1$

$f(b) = f(2) = 3(16) - 4(8)$

$= 48 - 32 = 16$

∴ Absolute maximum $= 16$

Absolute minimum $= -1$.

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(OR)

44 (b) $f(x,y) = \frac{3x}{y+\sin x}$

$f_x = \frac{(y+\sin x)(3) - (3x)(\cos x)}{(y+\sin x)^2}$

$= \frac{3y + 3\sin x - 3x\cos x}{(y+\sin x)^2}$

$f_{yx} = \frac{(y+\sin x)^2(3) - (3y+3\sin x - 3x\cos x)\,2(y+\sin x)(1)}{((y+\sin x)^2)^2}$

$= \frac{3(y+\sin x)^2 - 6(3y+3\sin x - 3x\cos x)(y+\sin x)}{(y+\sin x)^4}$

$= \frac{-3(y+\sin x)^2 + 6x\cos x(y+\sin x)}{(y+\sin x)^4}$ → (1)

$f_y = \frac{(y+\sin x)(0) - (3x)(1)}{(y+\sin x)^2}$

$= -\frac{3x}{(y+\sin x)^2}$

$f_{xy} = \frac{(y+\sin x)^2(-3) - (-3x)\,2(y+\sin x)(\cos x)}{((y+\sin x)^2)^2}$

$= \frac{-3(y+\sin x)^2 + 6x\cos x(y+\sin x)}{(y+\sin x)^4}$ → (2)

from (1), (2) $f_{xy} = f_{yx}$

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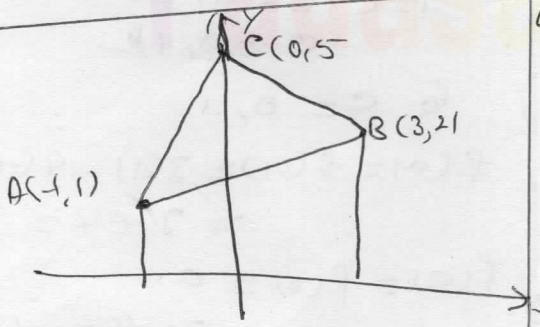
45 (a)




Equation of AB is $y = \frac{1}{4}(x+5)$

" BC is $y = -x+5$

" AC is $y = 4x+5$

$A = \int_{-1}^{0}(4x+5)\,dx + \int_{0}^{3}(-x+5)\,dx - \int_{-1}^{3}\frac{1}{4}(x+5)\,dx$ (VI)

$= \left(2x^2+5x\right]_{-1}^{0} + \left(\frac{-x^2}{2}+5x\right]_{0}^{3} - \frac{1}{4}\left(\frac{x^2}{2}+5x\right]_{-1}^{3}$

$= \frac{15}{2}$ Sq.units

---

45 (b) $y^2 - x^2\frac{dy}{dx} = xy\frac{dy}{dx}$

$y^2 = (x^2+xy)\frac{dy}{dx}$

$\therefore \frac{dy}{dx} = \frac{y^2}{x^2+xy}$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$v + x\frac{dv}{dx} = \frac{v^2x^2}{x^2+x(vx)} = \frac{v^2}{1+v}$

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$$x\frac{dv}{dx} = \frac{v^2}{1+v} - v$$

$$= \frac{v^2 + v - v^2}{1+v}$$

$$= \frac{-v}{1+v}$$

$$\left(\frac{1+v}{v}\right)dv = -\frac{dx}{x}$$

$$\left(1 + \frac{1}{v}\right)dv = -\frac{dx}{x}$$

$$\int\left(1 + \frac{1}{v}\right)dv = -\int\frac{dx}{x}$$

$$v + \log v = -\log x + c$$

$$\log v + \log x = -v + c$$

$$\log vx = -v + c$$

$$\log y = -\frac{y}{x} + c$$

$$e^{\log y} = e^{-\frac{y}{x} + c}$$

$$y = e^{-\frac{y}{x}} e^{c}$$

$$= e^{-\frac{y}{x}} \cdot c$$

46, (b)

| p | q | $\sim p$ | $p \wedge q$ | $p \vee q$ | $\sim(p \vee q)$ | $(\sim p) \wedge (\sim(p \vee q))$ |
|---|---|---|---|---|---|---|
| T | T | F | T | T | F | F |
| T | F | F | F | T | F | F |
| F | T | T | F | T | F | F |
| F | F | T | F | F | T | T |

$\therefore p \wedge q \equiv (p \vee q) \wedge (\sim p) \wedge (\sim(p \vee q))$

47 (a) $L_1 : \frac{x-6}{1} = \frac{y-2}{-2} = \frac{z-2}{2}$

$L_2 : \frac{x+4}{3} = \frac{y}{-2} = \frac{z-1}{-2}$

Distance $= D = \frac{|(\vec{c}-\vec{a}) \cdot (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|}$

$\vec{a} = 6\hat{i} + 2\hat{j} + 2\hat{k}$ | $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$

$\vec{c} = -4\hat{i} + 0\hat{j} + \hat{k}$ | $\vec{d} = 3\hat{i} - 2\hat{j} - 2\hat{k}$

$\vec{c} - \vec{a} = -10\hat{i} - 2\hat{j} - \hat{k}$

$$[\vec{c}-\vec{a}\ \ \vec{b}\ \ \vec{d}] = \begin{vmatrix} -10 & -2 & -1 \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix}$$

$$= -10(8) + 2(-8) - 1(4)$$

$$= -80 - 16 - 4$$

$$= -100 \neq 0$$

∴ The lines are skew. $|[\vec{c}-\vec{a}\ \ \vec{b}\ \ \vec{d}]| = 100$

$$\vec{b} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix}$$

$$= \hat{i}(8) - \hat{j}(-8) + \hat{k}(4)$$

$$= 4(2\hat{i} + 2\hat{j} + \hat{k})$$

$$|\vec{b} \times \vec{d}| = 4\sqrt{4+4+1} = 4 \times \sqrt{9} = 4 \times 3 = 12$$

$\therefore D = \frac{100}{12} = \frac{25}{3}$ units

(OR)

46 (a) P.M.F

(i)

| x | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|
| P(X=x) | 0.15 | 0.20 | 0.25 | 0.25 | 0.15 |

(ii) $P(X<1) = P(-1) + P(0)$

$= 0.15 + 0.20 = 0.35$

(iii) $P(X \geq 2) = P(2) + P(3)$

$= 0.25 + 0.15 = 0.40$

(OR)

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47) (b) Length of rectangle = $l$

Breadth of " = $b$

Area = $lb = K$

$b = \frac{K}{l}$

Perimeter $P = 2(l+b)$

$P(l) = 2\left(l + \frac{K}{l}\right)$

$P'(l) = 2\left(1 - \frac{K}{l^2}\right)$

Put $P'(l) = 0$

$2\left(1 - \frac{K}{l^2}\right) = 0$

$1 - \frac{K}{l^2} = 0$

$1 = \frac{K}{l^2}$

$l^2 = K \Rightarrow l = \sqrt{K}$

$P''(l) = 2 - K(-2/l^3)$

$= \frac{4K}{l^3}$

$P''(\sqrt{K}) = \frac{4K}{K\sqrt{K}} = \frac{4}{\sqrt{K}} > 0$

$\therefore l = \sqrt{K}$ is a minimum

$b = \frac{K}{\sqrt{K}} = \sqrt{K}$

$\therefore l = b = \sqrt{K}$.

$\therefore$ The given rectangle is a square.

Prepared by

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## PTA Model Question Paper – 4 Answer key (EM) 2019 - 2020

### Part – I

| Answer all the questions | |
|---|---|
| 1. | c) $\pm 3$ |
| 2. | c) 0 |
| 3. | a) $cis\frac{2\pi}{3}$ |
| 4. | c) n imaginary roots |
| 5. | b) $0 \leq x \leq \pi$ |
| 6. | c) -1 |
| 7. | c) an ellipse |
| 8. | c) 6 |
| 9. | a) 1 |
| 10. | c) $-\frac{1}{2}, -2$ |
| 11. | b) $\frac{1}{3}$ |
| 12. | d) 9 |
| 13. | b) -4 |
| 14. | b) $\frac{1}{(x+1)^2}dx$ |
| 15. | a) $y + \sin^{-1}x = c$ |
| 16. | b) $\frac{d^2y}{dx^2} + y = 0$ |
| 17. | b) $y = e^x + 2x + c$ |
| 18. | a) $1 \text{ and } \frac{1}{2}$ |
| 19. | d) $a * b = ab$ |
| 20. | c) both commutative and associative |

### Part – II

21. CQ
22. CQ
23. CQ
24. TB : 195 (Example: 5.23)
25. TB : 249 (Ex. 6.4– 5 (iii))
26. TB : 37 (Example 7.50)
27. CQ
28. TB : 148 (Ex. 10.1 – 1 (v))
29. TB : 261 (Ex. 11.3 – 6)
30. TB : 94 (Ex. 2.9 – 19)

### Part - III

31. CQ
32. TB : 101 (Example 3.2)
33. TB : 177 (Example 5.6)
34. TB : 231 (Ex. 6.1 -13)
35. TB : 44 (Ex. 7.7 – 2 (iii) )
36. CQ
37. CQ
38. TB : 113 (Ex. 9.3– 2 (iv) )

## PTA Model Question Paper – 4 Answer key (TM) 2019 - 2020

### பகுதி – I

| அனைத்து வினாக்களுக்கும் விடையளி | |
|---|---|
| 1. | c) $\pm 3$ |
| 2. | c) 0 |
| 3. | a) $cis\frac{2\pi}{3}$ |
| 4. | c) n கலப்பெண் மூலங்கள் |
| 5. | b) $0 \leq x \leq \pi$ |
| 6. | c) -1 |
| 7. | c) நீள்வட்டம் |
| 8. | c) 6 |
| 9. | a) 1 |
| 10. | c) $-\frac{1}{2}, -2$ |
| 11. | b) $\frac{1}{3}$ |
| 12. | d) 9 |
| 13. | b) -4 |
| 14. | b) $\frac{1}{(x+1)^2}dx$ |
| 15. | a) $y + \sin^{-1}x = c$ |
| 16. | b) $\frac{d^2y}{dx^2} + y = 0$ |
| 17. | b) $y = e^x + 2x + c$ |
| 18. | a) $1$ மற்றும் $\frac{1}{2}$ |
| 19. | d) $a * b = ab$ |
| 20. | c) பரிமாற்று விதி மற்றும் சேர்ப்பு விதிகளை நிறைவு செய்யாது |

### பகுதி - II

21. CQ
22. CQ
23. CQ
24. TB : 195 (எகா. 5.23)
25. TB : 249 (பயிற்சி. 6.4 – 5 (iii))
26. TB : 37 (எகா. 7.50)
27. CQ
28. TB : 148 (பயிற்சி 10.1 – 1 (v))
29. TB : 261 (பயிற்சி 11.3 – 6)
30. TB : 94 (பயிற்சி. 2.9 -19)

### பகுதி – III

31. CQ
32. TB : 101 (எ.கா. 3.2)
33. TB : 177 (எ.கா. 5.6)
34. TB : 231 (பயிற்சி. 6.1 -13)
35. TB : 44 (பயிற்சி 7.7 – 2(iii))
36. CQ
37. CQ
38. TB : 113 (பயிற்சி 9.3 -2(iv) )



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| | |
|---|---|
| 39. TB : 243 (Table -12.13)<br>40. CQ<br>**Part – IV**<br>41. a) TB : 47 (Ex. 1.7 - 2)<br>b) TB : 83 (Ex. 2.7 -5)<br>42. a) TB : 90 (Example. 2. 34)<br>b) CQ<br>43. a) CQ<br>b)TB : 266 (Ex. 6.8 – 4)<br>44. a) TB : 11 (Example: 7.14)<br>b) TB : 79 (Ex. 8.4 -6)<br>45. a) CQ<br>b)TB : 165 (Example: 10.21)<br>46. a)TB : 210 (Ex.11.4 - 7)<br>b) TB:234 (Example 12.9)<br>47. a) CQ<br>b) TB: 116 (Example9.36) | 39. TB : 243 (அட்டவணை -12.13)<br>40. CQ<br>**பகுதி - IV**<br>41. a) TB : 47 (பயிற்சி 1.7 - 2)<br>b) TB : 83 (பயிற்சி. 2.7 -5)<br>42. a) TB : 90 (எ.கா. 2. 34)<br>b) CQ<br>43. a) CQ<br>b)TB : 266 (பயிற்சி. 6.8 – 4)<br>44. a) TB : 11 (எ.கா. 7.14)<br>b) TB : 79 (பயிற்சி. 8.4 -6)<br>45. a) CQ<br>b)TB : 165 (எ.கா. 10.21)<br>46. a)TB : 210 (பயிற்சி.11.4 - 7)<br>b) TB:234 (எ.கா. 12.9)<br>47. a) CQ<br>b) TB: 116 (எ.கா. 9.36) |

CQ – Creative Questions



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## Answers

### Part – I

1, (c) $\pm 3$

2, (c) 0

3, (a) $\frac{2\pi}{3}$

4, (c) n imaginary roots.

5, (b) $0 \le x \le \pi$

6, (c) $-1$

7, (c) an ellipse.

8, (c) b

9, (a) 1

10, (c) $-\frac{1}{2}, -2$.

11, (b) $\frac{1}{3}$

12, (c) 9 (d)

13, (b) $-4$

14, (b) $\frac{1}{(x+1)^2}dx$

15, (a) $y + \sin^{-1}(x) = c$

16, (b) $\frac{d^2y}{dx^2} - y = 0$

17, (b) $y = e^x + 2x + c$.

18, (a) 1 and $\frac{1}{2}$

19, (d) $a * b = a^b$

20, (c) both commutative and associative

### Part – II

21, Let A be a non-singular matrix of order $2m+1$ where $m = 0, 1, 2 \ldots$

$\therefore |A| \neq 0$.

We know $|Adj\, A| = |A|^{n-1}$

22, $\tan^{-1}(\sin(-\frac{\pi}{2})) = \tan^{-1}(-\sin\frac{\pi}{2})$

$= \tan^{-1}(-1)$

$= \tan^{-1}(\tan(-\pi/4))$

$= -\frac{\pi}{4}$.

$\therefore$ Principal value is $-\frac{\pi}{4}$

23, There is only one $y^2$ in the given equation. the given equation represent a Parabola.

24, $\frac{x^2}{16} - \frac{y^2}{9} = 1$

$a^2 = 16 \quad b^2 = 9 \Rightarrow e = \frac{a^2+b^2}{a^2} = \frac{25}{16}$

$a = 4 \quad \Rightarrow e = \frac{5}{4}$

$ae = 5$

Foci $= (\pm ae, 0) = (\pm 5, 0)$

25, $\frac{x}{2} = \frac{y}{3} = \frac{2}{1}$; $\frac{x}{6} = \frac{y}{-1} = \frac{2}{-4}$.

$\vec{b} = 2\vec{i} + 3\vec{j} - \vec{k}$ $\vec{d} = 6\vec{i} - \vec{j} - 4\vec{k}$.

$\vec{b} \cdot \vec{d} = 12 - 3 + 4 = 13$

$|\vec{b}| = \sqrt{4+9+1} = \sqrt{14}$

$|\vec{d}| = \sqrt{36+1+16} = \sqrt{53}$

25, $\frac{x}{\frac{1}{2}} = \frac{y}{\frac{1}{3}} = \frac{2}{1}$; $\frac{x}{\frac{1}{6}} = \frac{y}{-1} = \frac{2}{-1/4}$

$\vec{b} = \frac{1}{2}\vec{i} + \frac{1}{3}\vec{j} - \vec{k}$ $\vec{d} = \frac{1}{6}\vec{i} - \vec{j} - \frac{1}{4}\vec{k}$

$\vec{b} \cdot \vec{d} = \frac{1}{12} - \frac{1}{3} + \frac{1}{4} = \frac{1-4+3}{12} = 0$

$\therefore \vec{b} \perp \vec{d} \Rightarrow \theta = \frac{\pi}{2}$

26, $f(x) = x^2 - 4x + 4$

$f'(x) = 2x - 4$

Put $f'(x) = 0 \Rightarrow 2x - 4 = 0$

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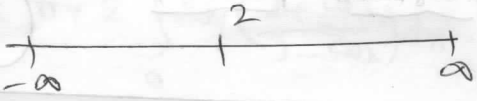



intervals are $(-\infty, 2)$ $(2, \infty)$

when $x < 2$ (say $x = 0$)

$f'(0) = -4 < 0$.

$\therefore$ $f$ is strictly decreasing function in $(-\infty, 2)$.

when $x > 2$ (say $x = 3$)

$f'(3) = 2 > 0$

$f$ is strictly increasing function in $(2, \infty)$.

---

27) $I = \int_0^1 \frac{(\sin^{-1}(x))^2}{\sqrt{1-x^2}}\,dx$

Put $u = \sin^{-1}x$

$du = \frac{1}{\sqrt{1-x^2}}\,dx$

when $x = 0$ $\quad u = \sin^{-1}(0) = 0$

$x = 1$ $\quad u = \sin^{-1}(1) = \frac{\pi}{2}$

$I = \int_0^{\pi/2} u^2\,du = \left[\frac{u^3}{3}\right]_0^{\pi/2}$

$= \frac{1}{3}\left(\frac{\pi^3}{8} - 0\right)$

$= \frac{\pi^3}{24}$ //

---

28) $y\left(\frac{dy}{dx}\right)^2 + 4\left(\frac{dy}{dx}\right)^4 = x$.

order $= 1$ ; degree $= 4$.

---

29) $f(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}(2x+1) & 0 \le x < 1 \\ 0 & x \ge 1 \end{cases}$

$\therefore f(x) = \begin{cases} \frac{1}{2}(2x+1); & 0 \le x < 1 \\ 0 & \text{otherwise} \end{cases}$

---

30) $(1+\omega)^7 = A + B\omega$

$(-\omega^2)^7 = A + B\omega$

$-\omega^{14} = A + B\omega$

$-\omega^2 = A + B\omega$

$1 + \omega = A + B\omega$

$\therefore A = 1, \; B = 1$

---

## Part – III

31) $A = \begin{bmatrix} 4 & -2 & 6 & 8 \\ 1 & 1 & -3 & -1 \\ 15 & -3 & 9 & 21 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 1 & -3 & -1 \\ 4 & -2 & 6 & 8 \\ 15 & -3 & 9 & 21 \end{bmatrix}$ $R_1 \leftrightarrow R_2$

$\sim \begin{bmatrix} 1 & 1 & -3 & -1 \\ 0 & -6 & 18 & 12 \\ 0 & -18 & 54 & 17 \end{bmatrix}$ $R_2 \to R_2 - 4R_1$, $R_3 \to R_3 - 15R_1$

$\sim \begin{bmatrix} 1 & 1 & -3 & -1 \\ 0 & -6 & 18 & 12 \\ 0 & 0 & 0 & -19 \end{bmatrix}$ $R_3 \to R_3 - 3R_2$

$\rho(A) = 3$

---

32) $\alpha + \beta = \frac{7}{2}$ $\quad \alpha\beta = \frac{13}{2}$

S.R $= \alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

$= \frac{49}{4} - 13 = \frac{49 - 52}{4}$

$= -\frac{3}{4}$

P.R $= \alpha^2\beta^2 = (\alpha\beta)^2 = \frac{169}{4}$

Required equation is

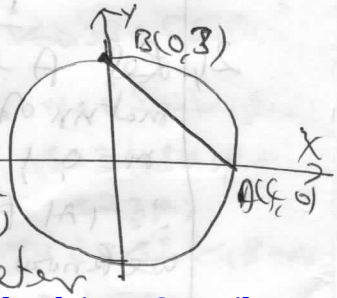
$x^2 - \left(-\frac{3}{4}\right)x + \frac{169}{4} = 0$

$4x^2 + 3x + 169 = 0$

---

33) $3x + 4y = 12$

$\frac{x}{4} + \frac{y}{3} = 1$




Equation of circle with A(4,0); B(0,3) as end of diameter

34) $\vec{F} = 3\vec{i} + 4\vec{j} - 5\vec{k}$.

$\vec{OP} = 4\vec{i} + 2\vec{j} - 3\vec{k}$

$\vec{OA} = 2\vec{i} - 3\vec{j} + 4\vec{k}$

$\vec{r} = \vec{OP} - \vec{OA} = 2\vec{i} + 5\vec{j} - 7\vec{k}$

moment $= \vec{r} \times \vec{F}$

$$\vec{r} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 5 & -7 \\ 3 & 4 & -5 \end{vmatrix}$$

$= \vec{i}(3) - \vec{j}(11) + \vec{k}(-7)$

$= 3\vec{i} - 11\vec{j} - 7\vec{k}$

magnitude $= \sqrt{9+121+49}$

$= \sqrt{179}$

D.c $= \left(\frac{3}{\sqrt{179}}; \frac{-11}{\sqrt{179}}; \frac{-7}{\sqrt{179}}\right)$.

35) $f(x) = x^2 e^{-2x}$

$f'(x) = x^2(-2e^{-2x}) + e^{-2x}(2x)$

$= -2x^2 e^{-2x} + 2xe^{-2x}$

$= 2xe^{-2x}(-x+1)$

Put $f'(x) = 0$

$2xe^{-2x}(-x+1) = 0$

$x = 0; \quad x = 1$

$f'(x) = 2e^{-2x}(-x^2+x)$

$f''(x) = 2\left(e^{-2x}(-2x+1) + (-x^2+x)(-2e^{-2x})\right)$

$= 2e^{-2x}(-2x+1+2x^2-2x)$

$= 2e^{-2x}(2x^2-4x+1)$

$f''(0) = 2(1)(1) = 2 > 0$

$x = 0$ is a minimum Point

$f''(1) = 2e^{-2}(-1) = -\frac{2}{e^2} < 0$

$x = 1$ is a maximum Point.

36) $r = 10.5$

$dr = 0.25$

$A = \pi r^2$

(i) $dA = 2\pi r\, dr$

$= 2\pi(10.5)(0.25)$

$= 21\pi(0.25)$

$= 5.25\pi$

ii) % error in A

$= \frac{dA}{A} \times 100$

$= \frac{5.25\pi}{\pi(10.5)^2} \times 100$

$= \frac{525}{(10.5)^2}$

$= \frac{525}{110.25} = 4.77\%$.

37) $u = e^{2(x-y)}$

$\log_e u = 2(x-y) = f(x,y)$

Degree $= 1$

$\therefore x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = 1 \times f$

$x\frac{\partial}{\partial x}(\log u) + y\frac{\partial}{\partial y}(\log u) = \log u$

$x\frac{1}{u}\frac{\partial u}{\partial x} + y\frac{1}{u}\frac{\partial u}{\partial y} = \log u$

$\therefore x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = u\log u$

38) $I = \int_0^{2\pi} x \log\left(\frac{3+\cos x}{3-\cos x}\right) dx$.

Let $f(x) = \log\left(\frac{3+\cos x}{3-\cos x}\right)$

$f(2\pi - x) = \log\left(\frac{3+\cos(2\pi-x)}{3-\cos(2\pi-x)}\right)$

$= \log\left(\frac{3+\cos x}{3-\cos x}\right) = f(x)$.

$\therefore I = \int_0^{2\pi} x \log\left(\frac{3+\cos x}{3-\cos x}\right) dx$

$= \frac{2\pi}{2}\int_0^{2\pi} \log\left(\frac{3+\cos x}{3-\cos x}\right) dx$

$= \pi \times 2 \int_0^{} \log\left(\frac{3+\cos x}{3-\cos x}\right) dx$.

$= 2\pi \times 0$ $\because$ If $f(2a-x) = -f(x)$

$= 0$ $\int_0^{2a} f(x)\,dx = 0$

39, (i)

| p | ~p | p ∨ ~p |
|---|---|---|
| T | F | T |
| F | T | T |

It is a tautology

(ii)

| p | ~p | p ∧ ~p |
|---|---|---|
| T | F | F |
| F | T | F |

It is a contradiction

40, $\frac{dP}{dt} = 5\% \text{ of } P$

$= 0.05P$

$P = e^{0.05t}$

for 2P t = ?

when $t = 0$ $P = 0$

Let P be the Population in t years and $P_0$ is the initial Population.

Given $\frac{dP}{dt} = 5\% \text{ of } P$

$= 0.05P$

$\therefore P = Ce^{0.05t}$

when $t = 0$, $P = P_0$

$P_0 = e^{(1)}$

$C = P_0$

$\therefore P = P_0 e^{0.05t}$

when $P = 2P_0$ $t = ?$

$2P_0 = P_0 e^{0.05t}$

$2 = e^{0.05t}$

$\log_e 2 = 0.05t$

$0.6931 = 0.05t$



41, (a) $\begin{pmatrix} 1 & 1 & 3 \\ 4 & 3 & \lambda \\ 2 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$[A/B] = \begin{bmatrix} 1 & 1 & 3 & 0 \\ 4 & 3 & \lambda & 0 \\ 2 & 1 & 2 & 0 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 1 & 3 & 0 \\ 2 & 1 & 2 & 0 \\ 4 & 3 & \lambda & 0 \end{bmatrix} R_1 \leftrightarrow R_3.$

$\sim \begin{bmatrix} 1 & 1 & 3 & 0 \\ 0 & -1 & -4 & 0 \\ 0 & -1 & \lambda-12 & 0 \end{bmatrix} R_2 \leftrightarrow R_2 - 2R_1$ $R_3 \leftrightarrow R_3 - 4R_1$

$\sim \begin{bmatrix} 1 & 1 & 3 & 0 \\ 0 & -1 & -4 & 0 \\ 0 & 0 & \lambda-8 & 0 \end{bmatrix} R_3 \leftrightarrow R_3 - R_2$

(i) If $\lambda \neq 8$ $\rho(A) = \rho[A, 0] = 3$.

The equations have unique solution.

(ii) If $\lambda = 8$ $\rho(A) = \rho[A, 0] = 2 <$ number of unknown

The equation have non-trivial solution.

(OR)

41, (b) $\cos\alpha + \cos\beta + \cos\gamma = 0$

$\sin\alpha + \sin\beta + \sin\gamma = 0$.

Let $a = \text{cis}\alpha$; $b = \text{cis}\beta$; $c = \text{cis}\gamma$

$a + b + c = \text{cis}\alpha + \text{cis}\beta + \text{cis}\gamma$

$= (\cos\alpha + \cos\beta + \cos\gamma) + i(\sin\alpha + \sin\beta + \sin\gamma)$

$= 0 + i(0)$

$= 0$

$a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$.

$(\text{cis}\alpha)^3 + (\text{cis}\beta)^3 + (\text{cis}\gamma)^3 = 3(\text{cis}\alpha)(\text{cis}\beta)(\text{cis}\gamma)$

$\cos 3\alpha + i\sin 3\alpha + \cos 3\beta + i\sin 3\beta + \cos 3\gamma + i\sin 3\gamma = 3[\cos(\alpha+\beta+\gamma) + i\sin(\alpha+\beta+\gamma)]$

$(\cos 3\alpha + \cos 3\beta + \cos 3\gamma) + i(\sin 3\alpha + \sin 3\beta + \sin 3\gamma) = 3\cos(\alpha+\beta+\gamma) + i\,3\sin(\alpha+\beta+\gamma)$.

Equating

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42 (a) $z^3 + 8i = 0$

$z^3 = -8i$

$= 8(-i)$

$= 8\left[\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)\right]$

$= 8\left[\cos\left(2k\pi - \frac{\pi}{2}\right) + i\sin\left(2k\pi - \frac{\pi}{2}\right)\right]$

$z = (8)^{\frac{1}{3}}\left[\cos\left(2k\pi - \frac{\pi}{2}\right) + i\sin\left(2k\pi - \frac{\pi}{2}\right)\right]^{\frac{1}{3}}$

$= 2\left[\cos\frac{1}{3}\left(2k\pi - \frac{\pi}{2}\right) + i\sin\frac{1}{3}\left(2k\pi - \frac{\pi}{2}\right)\right]$

$k = 0, 1, 2$

$= 2\left[\cos\left(\frac{4k\pi - \pi}{6}\right) + i\sin\left(\frac{4k\pi - \pi}{6}\right)\right]$

$k = 0, 1, 2$

$k=0,\ 2\left[\cos\left(\frac{-\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right]$

$= 2\left(\frac{\sqrt{3}}{2} - i\frac{1}{2}\right) = \sqrt{3} - i$

$k=1\quad 2\left[\cos\left(\frac{\pi}{2}\right) + i\sin\frac{\pi}{2}\right]$

$= 2(i) = 2i$

$k=2\quad 2\left[\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right]$

$= 2\left[\cos\left(\pi + \frac{\pi}{6}\right) + i\sin\left(\pi + \frac{\pi}{6}\right)\right]$

$= 2\left(-\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)$

$= 2\left(-\frac{\sqrt{3}}{2} - i\frac{1}{2}\right) = -\sqrt{3} - i$

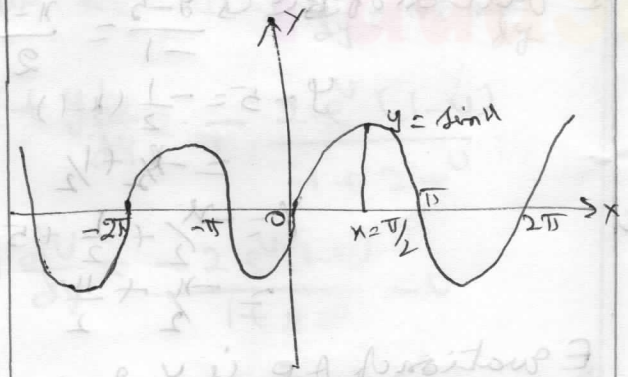
∴ The values of z are;

$\sqrt{3} - i;\ 2i;\ -\sqrt{3} - i$

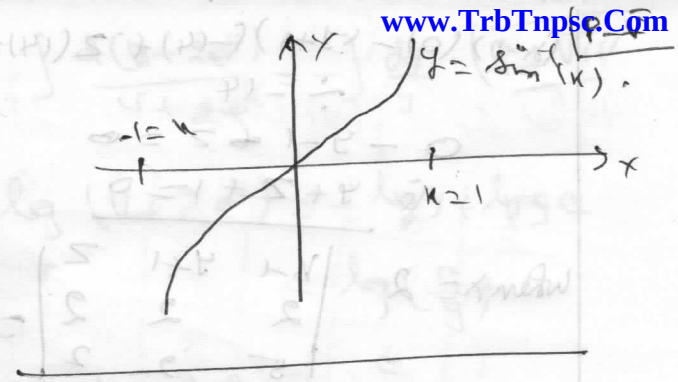
(OR)

42 (b)

| x | 0 | $\frac{\pi}{6}$ | $\pi/4$ | $\pi/3$ | $\frac{\pi}{2}$ | $\pi$ | $\frac{3\pi}{2}$ | $2\pi$ |
|---|---|---|---|---|---|---|---|---|
| $y = \sin x$ | 0 | $\frac{1}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1 | 0 | −1 | 0 |




| x | −1 | $-\frac{1}{\sqrt{2}}$ | 0 | $\frac{1}{\sqrt{2}}$ | 1 |
|---|---|---|---|---|---|
| $y = \sin^{-1}x$ | $-\frac{\pi}{2}$ | $-\frac{\pi}{4}$ | 0 | $\frac{\pi}{4}$ | $\frac{\pi}{2}$ |




43 (a) $e = \frac{1}{2}$

Centre = $C(0,0)$

Equation of directrix $x = 4$

$x = \frac{a}{e} = 4$

$a = 4e$

$= 4 \times \frac{1}{2} = 2$

$a^2 = 4.\quad b^2 = a^2(1 - e^2)$

$= 4(1 - 1/4)$

$= 4(3/4)$

$b^2 = 3$

Equation of an ellipse

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{x^2}{4} + \frac{y^2}{3} = 1$ (or) $3x^2 + 4y^2 = 12$

(OR)

43 (b) $(x_1, y_1, z_1) = (1, -1, 0)$

$(x_2, y_2, z_2) = (-1, -1, 0)$

$(b_1, b_2, b_3) = (2, \lambda, 2)$

$(d_1, d_2, d_3) = (5, 2, \lambda)$.

∵ the lines are coplanar;

$\begin{vmatrix} -2 & 0 & 0 \\ 2 & \lambda & 2 \\ 5 & 2 & \lambda \end{vmatrix} = 0$

$(-2)(\lambda^2 - 4) = 0 \Rightarrow \lambda^2 - 4 = 0$

$\lambda^2 = 4$

$\lambda = \pm 2$

Equation of the plane containing two lines $\begin{vmatrix} x-1 & y+1 & z \\ 2 & \lambda & 2 \\ 5 & 2 & \lambda \end{vmatrix} = 0$

when $\lambda = -2$ $\begin{vmatrix} x-1 & y+1 & z \\ 2 & -2 & 2 \\ 5 & 2 & -2 \end{vmatrix} = 0$

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$(x-1)(0) - (y+1)(-4) + z(4) = 0$

$\div -4$

$0 - y - 1 + z = 0$

$\Rightarrow y + z + 1 = 0$

When $x = 2$ $\begin{vmatrix} x-1 & y+1 & z \\ 2 & 2 & 2 \\ 5 & 2 & 2 \end{vmatrix} = 0$

$(x-1)(0) - (y+1)(-6) + z(-6) = 0$

$\div -6$

$0 - y - 1 + z = 0$

$y - z + 1 = 0$

---

44, (a) $y = x^2$ & $y = (x-3)^2$

$\therefore (x-3)^2 = x^2$

$x^2 - 6x + 9 = x^2$

$-2x + 3 = 0 \Rightarrow -2x = -3$

$x = 3/2$

$y = \frac{9}{4}$.

$\therefore$ Point of intersection $(\frac{3}{2}, \frac{9}{4})$

$y = x^2$

$\frac{dy}{dx} = 2x$

$m_1 = 2(3/2) = 3$

$y = (x-3)^2$

$\frac{dy}{dx} = 2(x-3)(1) = 2(x-3)$

$m_2 = 2(\frac{3}{2} - 3) = 2(-3/2)$

$m_2 = -3$

$\tan\theta = \left|\frac{3-(-3)}{1+(-9)}\right| = \left|\frac{6}{-8}\right|$

$= 3/4$

$\theta = \tan^{-1}(3/4)$

---

(OR)

44, (b) $W = \frac{1}{\sqrt{x^2+y^2+z^2}} = (x^2+y^2+z^2)^{-1/2}$

$W_x = -\frac{1}{2}(x^2+y^2+z^2)^{-3/2}(2x)$

$= -x(x^2+y^2+z^2)^{-3/2}$

$W_{xx} = (-x)\,-3/2\,(x^2+y^2+z^2)^{-5/2}(2x)$

$W_y = -\frac{1}{2}(x^2+y^2+z^2)^{-3/2}(2y)$

$= -y(x^2+y^2+z^2)^{-3/2}$

$W_{yy} = (-y)(-\frac{3}{2})(x^2+y^2+z^2)^{-5/2}(2y)$

$+ (x^2+y^2+z^2)^{-3/2}(-1)$

$= 3y^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$ → (2)

$W_{zz} = 3z^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$ → (3)

(1) + (2) + (3)

$W_{xx} + W_{yy} + W_{zz} = 3(x^2+y^2+z^2)^{-5/2}(x^2+y^2+z^2)$

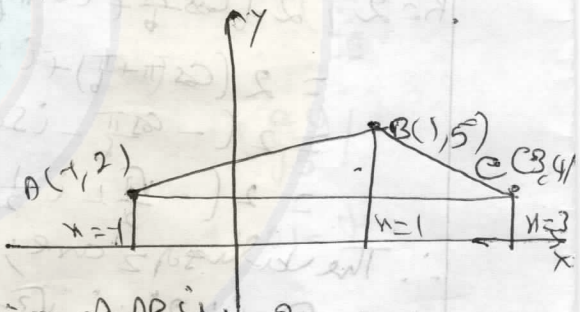
$- 3(x^2+y^2+z^2)^{-3/2}$

$= 3(x^2+y^2+z^2)^{-3/2} - 3(x^2+y^2+z^2)^{-3/2}$

$= 0$ //

---

45, (a) A(−1, 2) B(1, 5) & (3, 4).




Equation of AB is $\frac{y-2}{3} = \frac{x+1}{2}$

$y - 2 = \frac{3}{2}(x+1)$

$y = \frac{3}{2}x + \frac{3}{2} + 2$

$= \frac{3}{2}x + \frac{7}{2}$

Equation of BC is $\frac{y-5}{-1} = \frac{x-1}{2}$

$y - 5 = -\frac{1}{2}(x-1)$

$= -x/2 + 1/2$

$y = -\frac{x}{2} + \frac{1}{2} + 5$

$= -\frac{x}{2} + \frac{11}{2}$

Equation of AC is $\frac{y-2}{2} = \frac{x+1}{4}$

$y - 2 = \frac{1}{2}(x+1)$

$y - 2 = \frac{x}{2} + \frac{1}{2}$

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Area of $\triangle ABC$

$$= \int_{x=-1}^{x=1} \left(\frac{3}{2}x + \frac{7}{2}\right)dx + \int_{x=1}^{x=3} \left(-\frac{x}{2} + \frac{11}{2}\right)dx$$

$$= \int_{x=-1}^{x=3} \left(\frac{x}{2} + \frac{5}{2}\right)dx$$

$$= \left(\frac{3}{2}\frac{x^2}{2} + \frac{7}{2}x\right)_{-1}^{1} + \left(-\frac{1}{2}\frac{x^2}{2} + \frac{11}{2}x\right)_{1}^{3} - \left(\frac{1}{2}\frac{x^2}{2} + \frac{5}{2}x\right)_{-1}^{3}$$

$$= \left\{\left(\frac{3}{4} + \frac{7}{2}\right) - \left(\frac{3}{4} - \frac{7}{2}\right)\right\} + \left\{\left(-\frac{1}{4}(9) + \frac{33}{2}\right) - \left(-\frac{1}{4} + \frac{11}{2}\right)\right\} - \left\{\left(\frac{1}{4}(9) + \frac{15}{2}\right) - \left(\frac{1}{4} - \frac{5}{2}\right)\right\}$$

$$= \left(\frac{17}{4} + \frac{11}{4}\right) + \left\{\frac{57}{4} - \frac{21}{4}\right\} - \left\{\frac{39}{4} + \frac{9}{4}\right\}$$

$$= 7 + 9 - 12 = 16 - 12 = 4 \text{ Sq. units}$$

(OR)

45. (5) $(1 + 2e^{x/y})dx = -2e^{x/y}\left(1 - \frac{x}{y}\right)dy$

$$\frac{dx}{dy} = -\frac{2e^{x/y}\left(1 - \frac{x}{y}\right)}{1 + 2e^{x/y}}$$

Put $x = vy \Rightarrow \frac{dx}{dy} = v + y\frac{dv}{dy}$

$$v + y\frac{dv}{dy} = -\frac{2e^v(1 - v)}{1 + 2e^v}$$

$$y\frac{dv}{dy} = -\frac{2e^v(1 - v)}{1 + 2e^v} - v$$

$$= \frac{-2e^v + 2ve^v - v - 2ve^v}{1 + 2e^v}$$

$$= -\frac{(v + 2e^v)}{1 + 2e^v}$$

$$\int \frac{d(v + 2e^v)}{v + 2e^v} = -\int \frac{dy}{y} \quad (\text{P - VIII})$$

$$\log(v + 2e^v) = -\log y + \log c$$

$$= \log\left(\frac{c}{y}\right)$$

$$\therefore v + 2e^v = \frac{c}{y}$$

Substitute $v = \frac{x}{y}$

$$\frac{x}{y} + 2e^{\frac{x}{y}} = \frac{c}{y}$$

$$x + 2ye^{x/y} = c.$$

---

46. (a) $f(x) = \begin{cases} 16xe^{-4x} & x > 0 \\ 0 & x \le 0 \end{cases}$

$$\text{mean} = E(X) = \int_0^\infty x f(x)dx$$

$$= \int_0^\infty x(16xe^{-4x})dx$$

$$= 16\int_0^\infty x^2 e^{-4x}dx$$

By Gamma, $n = 2$ $a = 4$

$$= 16 \times \frac{\lfloor 2}{(4)^3} = 16 \times \frac{2}{16 \times 4}$$

$$\text{mean} = \frac{1}{2}$$

$$E(x^2) = \int_0^\infty x^2 f(x)dx$$

$$= \int_0^\infty x^2(16xe^{-4x})dx$$

$$= 16\int_0^\infty x^3 e^{-4x}dx$$

$n = 3$, $a = 4$

$$= 16\frac{\lfloor 3}{(4)^4} = 16 \times \frac{6}{16 \times 16}$$

$$E(x^2) = \frac{3}{8}$$

$$\text{Var}(X) = E(x^2) - [E(x)]^2$$

$$= \frac{3}{8} - \frac{1}{4} = \frac{3 - 2}{8}$$

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46) (b) $Z_5 = \{[0]\ [1]\ [2]\ [3]\ [4]\}$

| $+_5$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 2 | 3 | 4 |
| 1 | 1 | 2 | 3 | 4 | 0 |
| 2 | 2 | 3 | 4 | 0 | 1 |
| 3 | 3 | 4 | 0 | 1 | 2 |
| 4 | 4 | 0 | 1 | 2 | 3 |

(i) ∵ All entries are filled with elements of $Z_5$.
∴ $Z_5$ is closed under $+_5$

(ii) clearly $+_5$ is commutative

(iii) clearly $+_5$ is associative

(iv) Identity = 0.

(v) Inverse of 0 = 0
" 1 = 4
" 2 = 3
" 3 = 2
" 4 = 1.

---

47) (a) $\vec{b}\times\vec{c} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 0 \\ 0 & 1 & -1 \end{vmatrix}$

$= \vec{i}(-3) - \vec{j}(-2) + \vec{k}(2)$

$= -3\vec{i} + 2\vec{j} + 2\vec{k}$.

L.H.S $= \vec{a}\times(\vec{b}+\vec{c}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & -1 \\ -3 & 2 & 2 \end{vmatrix}$

$= \vec{i}(4) - \vec{j}(4) + \vec{k}(5)$

$= 4\vec{i} + \vec{j} + 5\vec{k}$.

$\vec{a}\cdot\vec{c} = 0+1+1 = 2$

$\vec{a}\cdot\vec{b} = 2+3+0 = 5$

R.H.S $= (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$

$= (2)(2\vec{i}+3\vec{j}) - (5)(\vec{j}-\vec{k})$

$= 4\vec{i} + 6\vec{j} - 5\vec{j} + 5\vec{k}$

$= 4\vec{i} + \vec{j} + 5\vec{k}$.

[L.H.S = R.H.S]

47) (b) Let $I = \int_0^{\pi/2} \frac{dx}{4\sin^2 x + 5\cos^2 x}$

$= \int_0^{\pi/2} \frac{\frac{1}{\cos^2 x}\,dx}{4\tan^2 x + 5}$

$= \int_0^{\pi/2} \frac{\sec^2 x\,dx}{5 + 4\tan^2 x}$

Put $u = \tan x$

$du = \sec^2 x\,dx$

when $x = 0$, $u = \tan 0 = 0$

$x = \frac{\pi}{2}$, $u = \tan\pi/2 = \infty$

$= \int_0^{\infty} \frac{du}{5 + 4u^2} = \int_0^{\infty} \frac{2\,du}{\frac{5}{4} + u^2}$

$= \frac{1}{4}\int_0^{\infty} \frac{du}{(\frac{\sqrt{5}}{2})^2 + u^2}$

$= \frac{1}{4}\left(\frac{1}{\frac{\sqrt{5}}{2}}\tan^{-1}\left(\frac{u}{\frac{\sqrt{5}}{2}}\right)\right]_0^{\infty}$

$= \frac{1}{4}\times\frac{2}{\sqrt{5}}\left[\tan^{-1}\left(\frac{2u}{\sqrt{5}}\right)\right]_0^{\infty}$

$= \frac{1}{2\sqrt{5}}\left[\tan^{-1}(\infty) - \tan^{-1}(0)\right]$

$= \frac{1}{2\sqrt{5}}\left(\frac{\pi}{2} - 0\right) = \frac{\pi}{4\sqrt{5}}$

---

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## PTA Model Question Paper – 5 Answer key (EM) 2019 - 2020

### Part – I

| Answer all the questions | |
|---|---|
| 1. | c) $\pm\frac{1}{3}$ |
| 2. | d) -1 |
| 3. | d) 0 |
| 4. | d) -1 |
| 5. | b) 1 |
| 6. | a) mn |
| 7. | d) 3 |
| 8. | d) $y = 1$ |
| 9. | c) $\frac{2}{\sqrt{3}}$ |
| 10. | c) 64 cubic units |
| 11. | d) $\frac{\pi}{2}$ |
| 12. | d) $\frac{1}{2}$ |
| 13. | c) (1,0) |
| 14. | b) $(1 + xy)e^{xy}$ |
| 15. | b) $2xu$ |
| 16. | b) $e^x + e^{-y} = c$ |
| 17. | b) $n, n + 1$ |
| 18. | d)16 *and* 24 |
| 19. | a) $\frac{11}{243}$ |
| 20. | c) 3 |

### Part – II

21. EX. 1.2 – 1 (iii)
22. EX. 2.4 -3
23. EX.4.2 - 2
24. CQ
25. Example 7. 36
26. Example 9.47
27. CQ
28. Ex. 11.4 – 1 (iii)
29. Ex. 12. 1– 4
30. CQ

### Part - III

31. CQ
32. CQ
33. Example 5.31
34. Ex. 6.3 - 7
35. Ex. 7.9– 2
36. CQ
37. Ex. 9.8– 2
38. Example. 10.9
39. Ex.11.5 - 8)
40. CQ

## PTA Model Question Paper – 5 Answer key (TM) 2019 - 2020

### பகுதி – I

| அனைத்து வினாக்களும் விடையளி | |
|---|---|
| 1. | c) $\pm\frac{1}{3}$ |
| 2. | d) -1 |
| 3. | d) 0 |
| 4. | d) -1 |
| 5. | b) 1 |
| 6. | a) m n |
| 7. | d) 3 |
| 8. | d) $y = 1$ |
| 9. | c) $\frac{2}{\sqrt{3}}$ |
| 10. | c) 64 கன அலகுகள் |
| 11. | d) $\frac{\pi}{2}$ |
| 12. | d) $\frac{1}{2}$ |
| 13. | c) (1,0) |
| 14. | b) $(1 + xy)e^{xy}$ |
| 15. | b) $2xu$ |
| 16. | b) $e^x + e^{-y} = c$ |
| 17. | b) $n, n + 1$ |
| 18. | d)16 மற்றும் 24 |
| 19. | a) $\frac{11}{243}$ |
| 20. | c) 3 |

### பகுதி - II

21. பயிற்சி. 1.2 – 1 (iii)
22. பயிற்சி. 2.4 -3
23. பயிற்சி.4.2 - 2
24. CQ
25. எ.கா 7. 36
26. எ.கா. 9.47
27. CQ
28. பயிற்சி. 11.4 – 1 (iii)
29. பயிற்சி. 12. 1– 4
30. CQ

### பகுதி – III

31. CQ
32. CQ
33. எ.கா 5.31
34. பயிற்சி. 6.3 - 7
35. பயிற்சி. 7.9– 2
36. CQ
37. பயிற்சி. 9.8– 2
38. TB : 156, (எ.கா. 10.9 )
39. பயிற்சி.11.5 - 8
40. CQ



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| Part – IV | பகுதி - IV |
|---|---|
| 41. a) Ex. 1.6 – 1 (iv)<br>b) Example. 2.35<br>42. a) Ex. 3. 2 - 4<br>b) Example 6.6<br>43. a)Ex. 6.8 – 1<br>b) CQ<br>44. a) Example: 7.17<br>b) CQ<br>45. a) Example: 9.27<br>b) Example: 10.15<br>46. a)CQ<br>b) Example 12.18<br>47. a) CQ<br>b) Ex. 5.1 -6 | 41. a)பயிற்சி. 1.6 – 1 (iv)<br>b) எ.கா. 2.35<br>42. a) பயிற்சி. 3. 2 - 4<br>b) எ.கா 6.6<br>43. a) பயிற்சி. 6.8 – 1<br>b)CQ<br>44. a) எ.கா: 7.17<br>b) CQ<br>45. a) எ.கா: 9.27<br>b)எ.கா : 10.15<br>46. a)CQ<br>b) எ.கா: 12.18<br>47. a) CQ<br>b) பயிற்சி. 5.1 -6 |



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# 12th Maths P.T.A Modal Question Paper-5 [P-I]

## Answer.

### Part-I

1. (c) $\pm \frac{1}{3}$
2. (b) $-1$.
3. (d) 0
4. (a) $-1$
5. (b) 1
6. (a) $mn$
7. (d) 3
8. (d) $y=1$
9. (c) $\frac{2}{\sqrt{3}}$
10. (c) 64 cubic units.
11. (d) $\frac{\pi}{2}$.
12. (d) $\frac{1}{2}$
13. (a) $(1,0)$
14. (b) $(1+xy)e^{xy}$
15. (b) 2 sq. units.
16. (b) $e^x + e^{-y} = c$
17. (b) $n; n+1$
18. (d) 16 and 24.
19. (a) $\frac{11}{243}$
20. (c) 3

### Part-II

21. $A = \begin{pmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & -2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \; R_2 \to R_2 - 3R_1$

$\rho(A) = 2$.

22. $\frac{z_1}{z_2} = \frac{2-i}{-4+3i} = \frac{2-i}{-4+3i} \times \frac{-4-3i}{-4-3i}$

$= \frac{(-8-3)+i(-6+4)}{16+9}$

$= \frac{-11+i(-2)}{25}$

$= \frac{-11}{25} + i\left(\frac{-2}{25}\right)$

22. $\left(\frac{z_1}{z_2}\right)^{-1} = \frac{1}{\frac{z_1}{z_2}} = \frac{z_2}{z_1} = \frac{-4+3i}{2-i}$

$= -\frac{4+3i}{2-i} \times \frac{2+i}{2+i}$

$= \frac{-11+i2}{5} = \frac{-11}{5} + i\frac{2}{5}$

23. $\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) = \cos^{-1}\left(\cos\frac{\pi}{6}\right) = \frac{\pi}{6} \neq -\frac{\pi}{6}$

$\therefore \cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \neq -\frac{\pi}{6}$

24. $\frac{y^2}{9} - \frac{x^2}{16} = 1$

$a^2 = 9 \Rightarrow a = 3$

$b^2 = 16$

Length of L.R $= \frac{2b^2}{a} = \frac{2(16)}{3}$

$= \frac{32}{3}$

25. $\lim_{x \to 0} \frac{\sin x}{x^2} = \frac{1}{0} = \infty$

$\lim_{x \to 0^+} \frac{\cos x}{2x} = \frac{1}{0} = \infty$.

$\lim_{x \to 0} \frac{\cos x}{2x} = -\infty$

Left and right limit are not same

$\therefore$ limit does not exist.

26. $A = \int_{x=-1}^{x=3} y\,dx$.

$5y = 30 - 6x \Rightarrow y = \frac{1}{5}(30-6x)$

$= \int_{-1}^{3} \frac{1}{5}(30-6x)\,dx$

$= \frac{1}{5}\left[30x - 3x^2\right]_{-1}^{3}$

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$= \frac{3}{5}\left[2+11\right] = \frac{3}{5} \times 32$

$= \frac{96}{5}$ Sq. units

27) $\frac{dy}{dx} + y = e^{-x}$

$P = 1 \quad Q = e^{-x}$

$\int P\,dx = \int 1\,dx = x$

I.F $= e^{x}$

Solution

$y e^{x} = \int e^{-x} e^{x}\,dx + c$

$= \int 1\,dx + c$

$y e^{x} = x + c$

28) mean $= E(x) = \int_1^2 x f(x)\,dx$

$= \int_1^2 x\, 2(x-1)\,dx$

$= 2\int_1^2 (x^2 - x)\,dx$

$= 2\left[\frac{x^3}{3} - \frac{x^2}{2}\right]_1^2$

$= 2\left\{\left(\frac{8}{3} - 2\right) - \left(\frac{1}{3} - \frac{1}{2}\right)\right\}$

$= 2\left[\frac{2}{3} + \frac{1}{6}\right]$

$= \frac{4}{3} + \frac{1}{3} = \frac{5}{3}$

29) Let $x = a + \sqrt{5}b$; $y = c + \sqrt{5}d$

$xy = (a+\sqrt{5}b)(c+\sqrt{5}d)$

$= (ac - 5bd) + \sqrt{5}(ad + bc)$

$\in A$.

∴ usual multiplication is a binary operation on A.

30) V.E of required plane

$\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

$\vec{a} = 2\vec{i} + 2\vec{j} + 3\vec{k}$

$\vec{n} = 3\vec{i} + 4\vec{j} + 3\vec{k}$.

$\vec{r} \cdot (3\vec{i} + 4\vec{j} + 3\vec{k}) = (2\vec{i} + 2\vec{j} + 3\vec{k}) \cdot (3\vec{i} + 4\vec{j} + 3\vec{k})$

## Part – III

31) $\begin{pmatrix} 5 & 2 \\ 7 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$

$AX = B \Rightarrow X = A^{-1}B$.

$|A| = 1 \quad \text{Adj } A = \begin{pmatrix} 3 & -2 \\ -7 & 5 \end{pmatrix}$

$A^{-1} = \frac{1}{1}\begin{pmatrix} 3 & -2 \\ -7 & 5 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ -7 & 5 \end{pmatrix}$

$X = A^{-1}B = \begin{pmatrix} 3 & -2 \\ -7 & 5 \end{pmatrix}\begin{pmatrix} 4 \\ 5 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$ $\quad x = 2 \quad y = -3$

32) Here the sum of odd and even Power are equal. ∴ $x = -1$ is a root

∴ $x + 1$ is a factor

∴ $2x^3 + 11x^2 - 9x - 18 = (x+1)(2x^2 + px - 18)$

Equating the coeff of $x$

$-9 = p - 18$

$p = 9$

∴ $2x^2 + 9x - 18 = 0$

$2x^2 + 12x - 3x - 18 = 0$

$2x(x+6) - 3(x+6) = 0$

$(2x-3)(x+6) = 0$

$2x - 3 = 0 \Rightarrow 2x = 3$

$x = 3/2$

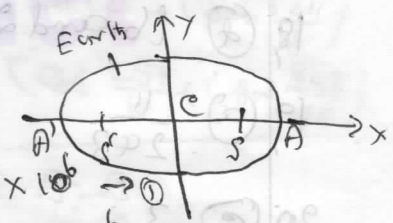
$x + 6 = 0 \Rightarrow x = -6$

∴ Solution set $= \{-1, -6, 3/2\}$

33) Given

$AS = a - ae$

$= 94.5 \times 10^6 \rightarrow (1)$




$A'S = a + ae = 152 \times 10^6 \rightarrow (2)$

(2) − (1) $2ae = 57.5 \times 10^6$

i.e $SS' = 575 \times 10^5$

The distance of the sun from other focus $= SS' = 575 \times 10^5$ km.

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34) $\vec{a}\times(\vec{b}\times\vec{c}) = l\vec{a}+m\vec{b}+n\vec{c}$

$(\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c} = l\vec{a}+m\vec{b}+n\vec{c}$

i.e. $(0)\vec{a} + (\vec{a}\cdot\vec{c})\vec{b} + (-(\vec{a}\cdot\vec{b}))\vec{c} = l\vec{a}+m\vec{b}+n\vec{c}$

$l = 0$

$m = \vec{a}\cdot\vec{c} = 3+4+3 = 10$

$n = -(\vec{a}\cdot\vec{b}) = -[2-2+3] = -3$

---

35) Let the positive numbers are $x, y$

Given $xy = 20 \Rightarrow y = \frac{20}{x}$

$u = x+y$ is minimum.

$u = x + \frac{20}{x}$

$u'(x) = 1 + 20(-1/x^2) = 1 - \frac{20}{x^2}$

Put $u' = 0 \Rightarrow 1 - \frac{20}{x^2} = 0$

$1 = \frac{20}{x^2}$

$x^2 = 20 = 4\times5$

$x = 2\sqrt{5}$

$u''(x) = 0 - 20(-2/x^3) = \frac{40}{x^3}$

$u''(2\sqrt{5}) = \frac{40}{(2\sqrt{5})^3} > 0$

$x = 2\sqrt{5}$ is a minimum point.

$\therefore y = \frac{20}{2\sqrt{5}} = \frac{4\times5}{2\times\sqrt{5}} = 2\sqrt{5}$

$\therefore x = y = 2\sqrt{5}$. Sum $= 4\sqrt{5}$

---

36) $f(x+\Delta x) = f(x) + f'(x)\Delta x$.

Let $f(x) = x^{1/5}$ and $x = 32$, $\Delta x = -1$

$\therefore x + \Delta x = 31$

$f(x) = (32)^{1/5} = (2^5)^{1/5} = 2$.

$f'(x) = \frac{1}{5}x^{-4/5} = \frac{1}{5x^{4/5}}$

$= \frac{1}{5(32)^{4/5}}$

$= \frac{1}{5(2^5)^{4/5}}$

$\therefore (31)^{1/5} = 2 + (0.0125)(-1)$

$= 2 - 0.0125$

$= 1.9875$

$\simeq 1.99$

---

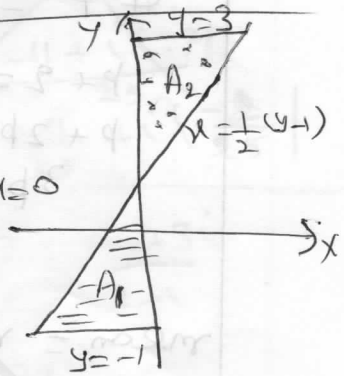
37) $2x = y-1$

$x = \frac{1}{2}(y-1)$

$\therefore$ Curve $y$-axis Put $x = 0$

$y - 1 = 0$

$y = 1$




$A = A_1 + A_2$

$= \int_{-1}^{1} (-x)\,dy + \int_{1}^{3} (x)\,dy$

$= -\int_{-1}^{1} \frac{1}{2}(y-1) + \int_{1}^{3} \frac{1}{2}(y+1)\,dy$

$= -\frac{1}{2}\left[\frac{y^2}{2} - y\right]_{-1}^{1} + \frac{1}{2}\left[\frac{y^2}{2} - y\right]_{1}^{3}$

$= -\frac{1}{2}\left\{\left(\frac{1}{2}-1\right) - \left(\frac{1}{2}+1\right)\right\} + \frac{1}{2}\left\{\left(\frac{9}{2}-3\right) - \left(\frac{1}{2}-1\right)\right\}$

$= \frac{1}{2}\left\{-\frac{1}{2} - \frac{3}{2}\right\} + \frac{1}{2}\left\{\frac{3}{2} + \frac{1}{2}\right\}$

$= \frac{1}{2}(-2) + \frac{1}{2}(2)$

$= 1+1 = 2$ Sq. units

---

38) $y = 2(x^2-1) + ce^{-x^2} \rightarrow (1)$

$\frac{dy}{dx} = 4x + ce^{-x^2}(-2x)$

$y_1 = 4x - (2xe^{-x^2})c$

$c(2xe^{-x^2}) = 4x - y_1$.

$ce^{-x^2} = \frac{1}{2x}(4x - y_1)$.

Substitute $c$ in (1)

$y = 2(x^2-1) + \frac{1}{2x}(4x - y_1)$

$2xy = 4x(x^2-1) + 4x - y_1$

$= 4x^3 - 4x + 4x - y_1$

$y_1 + 2xy - 4x^3 = 0$ is the required D.E

39) $n = 6$

$4P(X=4) = P(X=2)$

$4\left[{}^{6}C_4\, p^4 q^2\right] = {}^{6}C_2\, p^2 q^4$

$4\left[{}^{6}C_2\, p^4 q^2\right] = {}^{6}C_2\, p^2 q^4$

$4p^2 = q^2 \Rightarrow q = 2p$

$p + q = 1$

$p + 2p = 1$

$3p = 1 \Rightarrow p = \frac{1}{3}$

$q = \frac{2}{3}$

mean $= np = 6 \times \frac{1}{3} = 2.$

standard deviation $= \sqrt{npq}$

$= \sqrt{6 \times \frac{1}{3} \times \frac{2}{3}} = \frac{2}{\sqrt{3}}$

---

40) Given $x + iy = \sqrt{\frac{a+ib}{c+id}}$

$(x+iy)^2 = \frac{a+ib}{c+id}$

taking mod on both sides.

$|x+iy|^2 = \frac{|a+ib|}{|c+id|}$

$\left(\sqrt{x^2+y^2}\right)^2 = \frac{\sqrt{a^2+b^2}}{\sqrt{c^2+d^2}}$

i) $x^2+y^2 = \sqrt{\frac{a^2+b^2}{c^2+d^2}}$

---

## Part – IV

41) (a) $\begin{pmatrix} 2 & -1 & 1 \\ 6 & -3 & 3 \\ 4 & -2 & 2 \end{pmatrix} \begin{pmatrix} u \\ x \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix}$

$[A/B] = \begin{bmatrix} 2 & -1 & 1 & 2 \\ 6 & -3 & 3 & 6 \\ 4 & -2 & 2 & 4 \end{bmatrix}$

$\sim \begin{bmatrix} 2 & -1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $R_2 \leftrightarrow R_2 - 3R_1$, $R_3 \leftrightarrow R_3 - 2R_1$

$\rho(A) = \rho[AB] = 1 <$ no. of unknowns

The given system is consistent

These equation reduce to one equation

P-IV

$\begin{pmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$

$\therefore 2x - y + z = 2$

Put $y = s$; $z = t$.

$2x - s + t = 2 \Rightarrow 2x = 2 + s - t$

$x = \frac{1}{2}(2 + s - t)$

$\therefore$ S.S $= \left\{\frac{1}{2}(2+s-t); s; t\right\}; s, t \in R$

---

(OR)

41) (b) $\sqrt{3} + i = r(\cos\theta + i\sin\theta)$

$r\cos\theta = \sqrt{3}$

$r\sin\theta = 1$

$r^2 = 3 + 1 = 4 \Rightarrow r = 2$

$\left.\begin{aligned} \cos\theta &= \frac{\sqrt{3}}{2} \\ \sin\theta &= \frac{1}{2} \end{aligned}\right\} \Rightarrow \theta = \frac{\pi}{6}$

$\sqrt{3} + i = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

$= 2\left(\cos(2k\pi + \frac{\pi}{6}) + i\sin(2k\pi + \frac{\pi}{6})\right]$

$(\sqrt{3}+i)^{1/3} = 2^{1/3}\left(\cos(2k\pi + \frac{\pi}{6}) + i\sin(2k\pi + \frac{\pi}{6})\right)^{1/3}$

$= 2^{\frac{1}{3}}\left[\cos\frac{1}{3}(2k\pi + \frac{\pi}{6}) + i\sin\frac{1}{3}(2k\pi + \frac{\pi}{6})\right].$

$k = 0, 1, 2$

when $k = 0$ $\quad 2^{1/3}\left(\cos\frac{\pi}{18} + i\sin\frac{\pi}{18}\right)$

$k = 1$ $\quad 2^{1/3}\left(\cos\frac{13\pi}{18} + i\sin\frac{13\pi}{18}\right)$

$k = 2$ $\quad 2^{1/3}\left(\cos\frac{25\pi}{18} + i\sin\frac{25\pi}{18}\right).$

---

42) (a) one root $= \sqrt{5} - \sqrt{3}$

other root $= \sqrt{5} + \sqrt{3}$

S.R $= 2\sqrt{5}$

P.R $= (\sqrt{5}+\sqrt{3})(\sqrt{5}+\sqrt{3})$

$= 5 - 3 = 2$

one factor $x^2 - 2\sqrt{5}x + 2$

one root $= -\sqrt{5} - \sqrt{3}$

other root $= -\sqrt{5} + \sqrt{3}$

S.S $= \{\pm(\sqrt{5} \pm \sqrt{3})\}$

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Required equation

$(x^2-2\sqrt{5}x+2)(x^2+2\sqrt{5}x+2)=0$

$[(x^2+2)-2\sqrt{5}x][(x^2+2)+2\sqrt{5}x]=0$

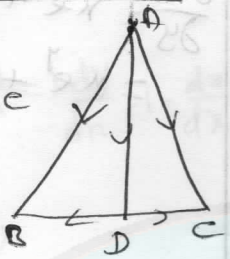
$(x^2+2)^2-20x^2=0$

$x^4+4x^2+4-20x^2=0$

$x^4-16x^2+4=0$

(OR)

42 (b)




$\because$ D is midpoint of BC

$\vec{BD} = \vec{DC}$

$\vec{C} - \vec{B} = \vec{DC}$

In $\triangle ADB$, $\vec{AB} = \vec{AD} + \vec{DB}$.

$(\vec{AB})^2 = (\vec{AD} - \vec{BD})^2$

$= AD^2 + BD^2 - 2\vec{AD}\cdot\vec{BD}$.

In $\triangle ADC$,

$\vec{AC} = \vec{AD} + \vec{DC}$.

~~$= \vec{AD} - \vec{BD}$.~~

$(\vec{AC})^2 = (\vec{AD} + \vec{DC})^2$

$= AD^2 + DC^2 + 2\vec{AD}\cdot\vec{DC}$

$= AD^2 + BD^2 + 2\vec{AD}\cdot\vec{BD}$

$AB^2 + AC^2 = 2(AB^2 + BD^2)$

(OR)

43 (a) Condition for coplanar lines

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$$

$(x_1, y_1, z_1) = (5, 7, -3)$ $\quad (b_1, b_2, b_3) = (4, 4, -5)$

$(x_2, y_2, z_2) = (8, 4, 5)$ $\quad (d_1, d_2, d_3) = (7, 1, 3)$

$$\begin{vmatrix} 3 & -3 & 8 \\ 4 & 4 & -5 \\ 7 & 1 & 3 \end{vmatrix} = 3(17) + 3(47) + 8(-24)$$

$= 51 + 141 - 192$

$= 192 - 192 = 0$

$\therefore$ the given lines are coplane.

Equation of the plane containing these lines. P-V

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$$

$$\begin{vmatrix} x-5 & y-7 & z+3 \\ 4 & 4 & -5 \\ 7 & 1 & 3 \end{vmatrix} = 0$$

$(x-5)(17) - (y-7)(47) + (z+3)(-24) = 0$

$17x - 85 - 47y + 329 - 24z - 72 = 0$

$17x - 47y - 24z + 172 = 0$

(OR)

(b)

$\vec{v}\cdot\vec{n} = \vec{r}\cdot(17\vec{i} - 47\vec{j} - 24\vec{k}) + 172 = 0$

(OR)

(b) Let r, h be the radius and height of a cylinder.

Volume $= V = \pi r^2 h = k$ (say)

$\therefore h = \frac{k}{\pi r^2}$.

Surface area $S = 2\pi r^2 + 2\pi r h$.

$= 2\pi r^2 + 2\pi r \times \frac{k}{\pi r^2}$

$S(r) = 2\pi r^2 + 2k\left(\frac{1}{r}\right)$

$S'(r) = 2\pi(2r) + 2k\left(-\frac{1}{r^2}\right)$

$= 4\pi r - \frac{2k}{r^2}$

Put $S'(r) = 0 \Rightarrow 4\pi r - \frac{2k}{r^2} = 0$

$2\pi r - \frac{k}{r^2} = 0$

$2\pi r = \frac{k}{r^2}$

$r^3 = \frac{k}{2\pi} \Rightarrow r = \left(\frac{k}{2\pi}\right)^{1/3}$

$S''(r) = 4\pi(1) - 2k(-2/r^3)$

$= 4\pi + \frac{4k}{r^3}$

$S''(r) = 4\pi + \frac{4k}{k} \times 2\pi$

$= 4\pi + 8\pi = 12\pi > 0$

$\therefore r = \left(\frac{k}{2\pi}\right)^{1/3}$ is the minimum point.

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44 (a) Let $P(x_1, y_1)$ is the point of intersection of the given curves.

$ax_1^2 + by_1^2 = 1 \rightarrow (1)$

$cx_1^2 + dy_1^2 = 1 \rightarrow (2)$

$(a-c)x_1^2 + (b-d)y_1^2 = 0$

$(a-c)x_1^2 = -(b-d)y_1^2$

$\frac{x_1^2}{y_1^2} = -\frac{(b-d)}{a-c}$.

$ax^2 + by^2 = 1$

$a(2x) + b(2yy_1) = 0$

$ax + byy_1 = 0$

$byy_1 = -ax$

$y_1 = -\frac{ax}{by}$

$m_1 = -\frac{ax_1}{by_1}$

$cx^2 + dy^2 = 1$

$c(2x) + d(2yy_1) = 0$

$cx + dyy_1 = 0$

$dyy_1 = -cx$

$y_1 = -\frac{cx}{dy}$

$m_2 = -\frac{cx_1}{dy_1}$

∵ they cut orthogonally

$m_1 m_2 = -1$

$\left(-\frac{ax_1}{by_1}\right)\left(-\frac{cx_1}{dy_1}\right) = -1$

$\frac{ac}{bd} \times \frac{x_1^2}{y_1^2} = -1$

$\frac{ac}{bd} \times -\frac{(b-d)}{a-c} = -1$

$\frac{b-d}{bd} = \frac{a-c}{ac}$

$\frac{1}{d} - \frac{1}{b} = \frac{1}{c} - \frac{1}{a}$

$\therefore \frac{1}{a} - \frac{1}{b} = \frac{1}{c} - \frac{1}{d}$

(OR)

44 (b) $z = xe^y + ye^{-x}$

$\frac{\partial z}{\partial x} = e^y(1) + y(-e^{-x})$

$= e^y - ye^{-x}$

$\frac{\partial z}{\partial y} = xe^y + e^{-x}(1)$

$= xe^y + e^{-x}$

$x = e^{-t}$

$\frac{dx}{dt} = -e^{-t}$

$\frac{\partial x}{\partial s} = 0$

$y = st^2$

$\frac{\partial y}{\partial t} = s(2t) = 2st$

$\frac{\partial y}{\partial s} = t^2(1) = t^2$.

(i) $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$

$= (e^y - ye^{-x})(0) + (xe^y + e^{-x})t^2$

$= 0 + (xe^y + e^{-x})t^2$

$= (e^{-t}e^{st^2} + e^{-t})t^2$

(ii) $\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$

$= (e^y - ye^{-x})(-e^{-t}) + (xe^y + e^{-x})(2st)$

$= (e^{st^2} - st^2e^{-t})(-e^{-t})$

$+ (e^{-t}e^{st^2} + e^{-e^{-t}})(2st)$

45 (a) Let $I = \int_0^{\pi/4} \log(1+\tan x)\,dx \rightarrow (1)$

$I = \int_0^{\pi/4} \log\left[1 + \tan\left(\frac{\pi}{4} - x\right)\right] dx$

$= \int_0^{\pi/4} \log\left[1 + \frac{1-\tan x}{1+\tan x}\right] dx$

$= \int_0^{\pi/4} \log\left(\frac{2}{1+\tan x}\right) dx \rightarrow (2)$

(1) + (2)

$I + I = \int_0^{\pi/4} \log(1+\tan x)\,dx + \int_0^{\pi/4} \log\left(\frac{2}{1+\tan x}\right)$

$= \int_0^{\pi/4} \log\left[(1+\tan x) \times \frac{2}{1+\tan x}\right]$

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$= \log 2 \,[x]_0^{\frac{\pi}{4}}$

$= (\log 2)\frac{\pi}{4}$

$I = (\log 2)\frac{\pi}{8}$

(OR)

45 ⑤ Put $z = x - y$

$\frac{dz}{dx} = 1 - \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = 1 - \frac{dz}{dx}$

$1 - \frac{dz}{dx} = \frac{z+5}{2z+7}$

$1 - \frac{z+5}{2z+7} = \frac{dz}{dx}$

$\frac{dz}{dx} = \frac{2z+7-z-5}{2z+7}$

$= \frac{z+2}{2z+7}$

$\left(\frac{2z+7}{z+2}\right) dz = dx$

$\left(\frac{2z+4+3}{z+2}\right) dz = dx$

$\left(\frac{2(z+2)+3}{z+2}\right) dz = dx$

$\left[2 + 3\,\frac{1}{z+2}\right] dz = dx$

$2\int dz + 3\int \frac{dz}{z+2} = \int dx$

$2z + 3\log(z+2) = x + c$

$2(x-y) + 3\log(x-y+2) = x + c$

46 ⓐ $n = 5$

$np + npq = 4.8$

$np(1+q) = 4.8$

$5p(1+1-p) = 4.8$

$p(2-p) = \frac{4.8}{5}$

$2p - p^2 = 0.96$

(i) $p^2 - 2p + 0.96 = 0$

$\because p \neq 1 \quad p = 0.8$

$q = 0.2$

$P(X = x) = {}^5C_x (0.8)^x (0.2)^{5-x}$;

$x = 0, 1, 2 \ldots 5.$

46 ⓑ

| p | q | $p \leftrightarrow q$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | F |
| F | F | T |

| p | q | $p \to q$ | $q \to p$ | $(p \to q) \wedge (q \to p)$ |
|---|---|---|---|---|
| T | T | T | T | T |
| T | F | F | T | F |
| F | T | T | F | F |
| F | F | T | T | T |

The last column of the table are same.

$\therefore p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

47 ⓐ Let $\theta = \tan^{-1}(x)$

$\tan\theta = x = \frac{x}{1}$

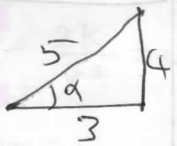
$\cos\theta = \frac{1}{\sqrt{1+x^2}} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$

L.H.S $= \cos(\tan^{-1}(x))$

$= \cos\theta = \frac{1}{\sqrt{1+x^2}}$

Let $\alpha = \cot^{-1}(3/4)$

$\cot\alpha = 3/4$




$\sin\alpha = \frac{3}{5}$

R.H.S $= \sin(\cot^{-1}(3/4))$

$= \sin\alpha.$

$= 3/5.$

Now $\cos(\tan^{-1}(x)) = \sin(\cot^{-1}(3/4))$

$\frac{1}{\sqrt{1+x^2}} = \frac{3}{5}$

$\frac{5}{3} = \sqrt{1+x^2}$

$x^2 = \frac{25}{9} - 1$

$= \frac{16}{9}$

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(OR)

47. (5) Let the equation of required circle is

$x^2+y^2+2gx+2fy+c=0$.

∵ it passes through (1, 0)

$1+2g+c=0$

$2g+c=-1 \rightarrow (1)$

∵ it passes through (-1, 0)

$1-2g+c=0$

$-2g+c=-1 \rightarrow (2)$

∵ it passes through (0, 1)

$1+2f+c=0$

$2f+c=-1 \rightarrow (3)$

(1) + (2) $2c=-2$

$c=-1$

$2g-1=-1$

$2g=0 \Rightarrow g=0$

$2f-1=-1$

$2f=0 \Rightarrow f=0$

∴ Equation of the circle

$x^2+y^2-1=0$

$x^2+y^2=1$

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P. G. Maths

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## PTA Model Question Paper – 6 Answer key (EM) 2019 - 2020

### Part – I

| Answer all the questions | |
|---|---|
| 1. | b)consistent |
| 2. | c) $adj\ (adj\ A) = \|A\|^n.\|A\|$ |
| 3. | c)2 |
| 4. | c) 1 |
| 5. | a) $\frac{\pi}{2}$ |
| 6. | b) $2\ (a^2 + b^2)$ |
| 7. | a) $\frac{1}{\sqrt{2}}$ |
| 8. | b) $\frac{3\pi}{4}$ |
| 9. | c) $x + 2y - 2z = 1$ |
| 10. | a) 1 |
| 11. | c)$\frac{\sqrt{3}}{12}$ |
| 12. | d) 4 |
| 13. | a) $z - x$ |
| 14. | c) $x \cos x$ |
| 15. | a) 0 |
| 16. | a) 2; 3 |
| 17. | c) parabola |
| 18. | d) 2 |
| 19. | a) 1 |
| 20. | c) N |

### Part – II

21. Exercise 1.1 - 9
22. CQ
23. Exercise 5.1 – 11(iv)
24. Example: 6.48
25. CQ
26. CQ
27. Example: 10.5
28. Exercise 11.5 -1(ii)
29. Example: 12.12
30. CQ

### Part - III

31. CQ
32. CQ
33. Example 5.22
34. Exercise 6.5 -3
35. CQ
36. CQ
37. Exercise 10.5 -4 (i)
38. CQ
39. Example 10.20
40. CQ

## PTA Model Question Paper – 6 Answer key (TM) 2019 - 2020

### பகுதி – I

| அனைத்து வினாக்களுக்கும் விடையளி | |
|---|---|
| 1. | b)ஒருங்கமைவுடையது |
| 2. | c) $adj\ (adj\ A) = \|A\|^n.\|A\|$ |
| 3. | c)2 |
| 4. | c) 1 |
| 5. | a) $\frac{\pi}{2}$ |
| 6. | b) $2\ (a^2 + b^2)$ |
| 7. | a) $\frac{1}{\sqrt{2}}$ |
| 8. | b) $\frac{3\pi}{4}$ |
| 9. | c) $x + 2y - 2z = 1$ |
| 10. | a) 1 |
| 11. | c)$\frac{\sqrt{3}}{\sqrt{2}}$ |
| 12. | d) 4 |
| 13. | a) $z - x$ |
| 14. | c) $x \cos x$ |
| 15. | a) 0 |
| 16. | a) 2; 3 |
| 17. | c) பரவளையம் |
| 18. | d) 2 |
| 19. | a) 1 |
| 20. | c) N |

### பகுதி - II

21. பயிற்சி 1.1 - 9
22. CQ
23. பயிற்சி 5.1 – 11(iv)
24. எ.கா: 6.28
25. CQ
26. CQ
27. எ.கா: 10.5
28. பயிற்சி 11.5 -1(ii)
29. எ.கா: 12.12
30. CQ

### பகுதி – III

31. CQ
32. CQ
33. எ.கா 5.22
34. பயிற்சி 6.5 -3
35. CQ
36. CQ
37. பயிற்சி 10.5 -4 (i)
38. CQ
39. எ.கா 10.20
40. CQ



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| Part – IV | பகுதி - IV |
|---|---|
| 41. a) Example. 1.21<br>b) Example. 2.27<br>42. a) Exercise 3.5-7<br>b) Example. 4.27<br>43. a) CQ<br>b) Example. 6.16<br>44. a) Exercise 6.5-5<br>b) CQ<br>45. a) Exercise 7.5-12<br>b) CQ<br>46. a) Exercise 9.8-9<br>b) Exercise 12.2-15<br>47. a) CQ<br>b) Exercise 11.4-4 | 41. a) எ.கா. 1.21<br>b) எ.கா. 2.27<br>42. a) பயிற்சி 3.5-7<br>b) எ.கா. 4.27<br>43. a) CQ<br>b) எ.கா. 6.16<br>44. a) பயிற்சி 6.5-5<br>b) CQ<br>45. a) பயிற்சி 7.5-12<br>b) CQ<br>46. a) பயிற்சி 9.8-9<br>b) பயிற்சி 12.2-15<br>47. a) CQ<br>b) பயிற்சி 11.4-4 |

CQ – Creative Question



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12th Maths P.T.A Modal Question Paper-6 P-I

Answers

Part - I

1. (b) Consistent
2. (c) $adj(adjA) = |A|^n \cdot |A|$
3. (c) 2
4. (c) 1
5. (a) $\frac{\pi}{2}$
6. (b) $2(a^2+b^2)$
7. (a) $\frac{1}{\sqrt{2}}$
8. (b) $\frac{3\pi}{4}$
9. (c) $x+2y-2z=1$
10. (a) 1
11. (c) $\frac{\sqrt{3}}{2}$
12. (d) 4.
13. (a) $2-x$
14. (c) $x\cos x$
15. (a) 0
16. (a) 2; 3
17. (c) parabola
18. (d) 2
19. (a) 1
20. (c) N

Part - II

21. $Adj A = \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$

$|Adj A| = 2(18) = 36$

$A^{-1} = \frac{\pm 1}{\sqrt{36}}(adj A)$

22. $-1+i\sqrt{3} = r(\cos\theta + i\sin\theta)$

$r\cos\theta = -1$

$r\sin\theta = \sqrt{3}$

$r^2 = 1+3 = 4 \Rightarrow r = 2$

$\cos\theta = \frac{-1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$ $\Rightarrow \theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

$-1+i\sqrt{3} = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$

$= 2\left[\cos\left(2k\pi + \frac{2\pi}{3}\right) + i\sin\left(2k\pi + \frac{2\pi}{3}\right)\right]$;

$k = 0, 1, 2.$

23. $x^2+y^2-3x+2y+1=0$

$2g = -3$ | $2f = 2$ | $c = 1$

$g = -3/2$ | $f = 1$

Centre $= C(3/2, -1)$

Radius $= \sqrt{\frac{9}{4}+1-1} = \sqrt{\frac{9}{4}} = \frac{3}{2}$

24. $\sin\theta = \frac{\vec{b}\cdot\vec{n}}{|\vec{b}||\vec{n}|}$

$\vec{b} = \vec{i}-\vec{j}+\vec{k}$ $\vec{n} = 2\vec{i}-\vec{j}+\vec{k}$

$\vec{b}\cdot\vec{n} = 2+1+1 = 4$

$|\vec{b}| = \sqrt{3}$ $|\vec{n}| = \sqrt{6}$

$\sin\theta = \frac{4}{\sqrt{3}\sqrt{6}} = \frac{4}{\sqrt{3}\times\sqrt{2}\times\sqrt{3}}$

$= \frac{2\times 2\sqrt{2}}{\sqrt{2}\times 3}$

$= \frac{2\sqrt{2}}{3}$

$\theta = \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

25. $f(x) = |\frac{1}{x}|$; $x \in [-1, 1]$

$f(x)$ is continuous in $[-1, 1]$

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26) $\because x=0$ to $x=1$, $|x|=x$

$$\therefore \int_0^1 \frac{|x|}{x}\,dx = \int_0^1 \frac{x}{x}\,dx$$

$$= \int_0^1 dx = (x)_0^1$$

$$= 1-0$$

$$= 1$$

27) $y^2 = 4ax$

$2yy_1 = 4a(1)$

$= 4a$

$yy_1 = 2a \Rightarrow a = \frac{yy_1}{2}$

D.E is $y^2 = 4\left(\frac{yy_1}{2}\right)x$

$= 2(yxy_1)$

$y = 2xy_1$.

(or) $2xy_1 - y = 0$

28) $n=10$, $p=\frac{1}{5} \Rightarrow q=\frac{4}{5}$

$k=4$

$P(x=k) = {}^nC_k\, p^k\, q^{n-k}$

$P(x=4) = {}^{10}C_4 \left(\frac{1}{5}\right)^4 \left(\frac{4}{5}\right)^6$

29) p: it is cold ~p: it is not cold.

q ∨ ~p : it is raining (or) it is not cold.

30) $\cos^{-1}\left(\frac{1}{2}\right) - 2\sin^{-1}\left(-\frac{1}{2}\right)$

$= \frac{\pi}{3} - 2\sin^{-1}\left(\sin\left(-\frac{\pi}{6}\right)\right)$

$= \cos^{-1}\left(\cos\frac{\pi}{3}\right) - 2\sin^{-1}\left(\sin\left(-\frac{\pi}{6}\right)\right)$

$= \frac{\pi}{3} - 2\left(-\frac{\pi}{6}\right)$

$= \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$

## Part – III

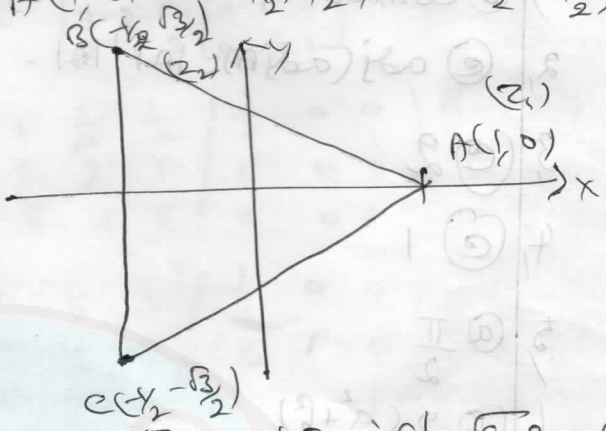
31) $A = \begin{pmatrix} 1 & 3 \\ 2 & -5 \end{pmatrix}$ $|A| = -11$

Adj $A = \begin{pmatrix} -5 & -3 \\ -2 & 1 \end{pmatrix}$

$|A|I = -11\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -11 & 0 \\ 0 & -11 \end{pmatrix}$

$\therefore A(\text{adj}A) = (\text{adj}A)A = |A|I$

32) A(1, 0) B$\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ C$\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$




$AB = |z_1 - z_2| = \left|\frac{3}{2} - i\frac{\sqrt{3}}{2}\right| = \sqrt{\frac{9}{4}+\frac{3}{4}} = \sqrt{3}$

$BC = |z_2 - z_3| = \left|2i\frac{\sqrt{3}}{2}\right| = |i\sqrt{3}| = \sqrt{}$

$CA = |z_3 - z_1| = \left|-\frac{3}{2} - i\frac{\sqrt{3}}{2}\right| = \sqrt{\frac{9}{4}+\frac{3}{4}} =$

$\therefore AB = BC = CA = \sqrt{3}$

$\therefore$ ABC is an equilateral tria

33) Vertices = A$(0, \pm 4) = (0, \pm a)$

Foci = f$(0, \pm 6) = (0, \pm ae)$

$a = 4$ $ae = 6$

$a^2 = 16$

$b^2 = (ae)^2 - a^2 = 36 - 16 = 20$

Equation of the hyperbola

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$\frac{y^2}{16} - \frac{x^2}{20} = 1 \text{ (or) } 5y^2 - 4x^2 = 80$$

34) $L_1 : \frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4} = t$

$\therefore P(2t+1, 3t-1, 4t+1)$

$L_2 : \frac{x-3}{1} = \frac{y-m}{2} = \frac{z-0}{1} = s$

$Q(s+3, 2s+m, s)$

$4t+1 = 0$ | $2t+1 = s+3$

$4t = -1$ | $-\frac{1}{2}+1 = s+3$

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$2\delta + m = 3t - 1$

$-5 + m = -\frac{3}{4} - 1$

$= -\frac{7}{4}$

$m = -\frac{7}{4} + 5$

$= \frac{-7+20}{4}$

$m = \frac{13}{4}$

34, Condition for intersecting lines

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$$

$(x_1 y_1 z_1) = (1, -1, 1)$ $(x_2 y_2 z_2) = (3, m, 0)$

$(b_1 b_2 b_3) = (2, 3, 4)$ $(d_1 d_2 d_3) = (1, 2, 1)$

$$\begin{vmatrix} 2 & m+1 & -1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$2(-5) - (m+1)(-2) - 1(1) = 0$

$-10 + 2m + 2 - 1 = 0$

$2m - 9 = 0$

$2m = 9 \Rightarrow m = \frac{9}{2}$

35, $w = xy + \sin(xy)$

$w_x = y(1) + \cos(xy)(y)$

$= y + y\cos(xy)$

$w_{yx} = 1 + \cos(xy)(1) + y(-\sin(xy))(x)$

$= 1 + \cos(xy) - xy\sin(xy) \rightarrow (1)$

$w_y = x(1) + \cos(xy)(x)$

$= x + x\cos(xy)$

$w_{xy} = 1 + \cos(xy) + x(-\sin(xy))(y)$

$= 1 + \cos(xy) - xy\sin(xy) \rightarrow (2)$

$\therefore w_{yx} = w_{xy}$

36, $I = \int_0^{\pi/3} \frac{\sec x \tan x \, dx}{1 + (\sec x)^2}$ P-III

Put $u = \sec x \Rightarrow du = \sec x \tan x \, dx$.

when $x = 0$ $u = 1$

$x = \frac{\pi}{3}$ $u = \sec\frac{\pi}{3} = 2$.

$= \int_1^2 \frac{du}{1+u^2}$

$= \left[\tan^{-1}(u)\right]_1^2$

$= \tan^{-1}(2) - \tan^{-1}(1)$

$= \tan^{-1}(2) - \pi/4$

37) $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$

$\int \frac{dy}{\sqrt{1-y^2}} = \int \frac{dx}{\sqrt{1-x^2}}$

$\sin^{-1}(y) = \sin^{-1}(x) + c$

38, $y = e^{3x}(c\cos 2x + D\sin 2x)$

$ye^{-3x} = c\cos 2x + D\sin 2x \rightarrow (1)$

Diff w.r.t x

$y(-3e^{-3x}) + e^{-3x}y_1 = c(-2\sin 2x) + D(2\cos 2x)$

$(y_1 - 3y)e^{-3x} = -2c\sin 2x + 2D\cos 2x$

Again diff w.r.t x

$(y_1 - 3y)(-3e^{-3x}) + e^{-3x}(y_2 - 3y_1)$

$= -2c(2\cos 2x) + 2D(-2\sin 2x)$

$= -4(c\cos 2x + D\sin 2x)$

$= -4ye^{-3x}$ [from (1)]

$\therefore e^{-3x}$

$(y_1 - 3y)(-3) + (y_2 - 3y_1) = -4y$

$-3y_1 + 9y + y_2 - 3y_1 = -4y$

$y_2 - 6y_1 + 13y = 0$

(OR)

$\alpha = 3$ $\beta = 2$ $m_1 = \alpha + i\beta = 3 + 2i$

$m_2 = \alpha - i\beta = 3 - 2i$

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39, $y^2 + x^2\frac{dy}{dx} = xy\frac{dy}{dx}$

$y^2 = (xy - x^2)\frac{dy}{dx}$

$\therefore \frac{dx}{dy} = \frac{xy - x^2}{y^2}$

Put $x = vy \Rightarrow \frac{dv}{dy} = v + y\frac{dv}{dy}$

$v + y\frac{dv}{dy} = \frac{vy^2 - v^2y^2}{y^2}$

$v + y\frac{dv}{dy} = v - v^2$

$y\frac{dv}{dy} = -v^2$

$\frac{dv}{v^2} = -\frac{dy}{y}$

$\int v^{-2}\,dv = -\int \frac{dy}{y}$

$\frac{v^{-1}}{-1} = -\log y + c$

$\log y = \frac{1}{v} + c$

$\log y = \frac{x}{y} + c$

$y = e^{x/y + c}$

$= e^{x/y} e^{c}$

$= k e^{x/y}$

---

40, when $x = 0$ $\quad y^3 - 8 = 0$

$y^3 = 8$

$y = 2$

Required Point $= (0, 2)$

$x^2y - x = y^3 - 8$

$x^2y_1 + y(2x) - 1 = 3y^2y_1 - 0$

$x^2y_1 - 3y^2y_1 = 1 - 2xy$

$(x^2 - 3y^2)y_1 = 1 - 2xy$

$y_1 = \frac{1 - 2xy}{x^2 - 3y^2}$

$m = (y_1)_{at\,(0,2)} = \frac{1 - 0}{0 - 3(4)}$

$= \frac{1}{-12}$

$m = -\frac{1}{12}$

Equation of tangent at $(0,2)$ is

$y - 2 = -\frac{1}{12}(x - 0)$

Part - IV

41, (a) $A = \begin{pmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix}$

$[A/I_3] = \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 3 & 2 & 1 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$ $R_1 \to \frac{1}{2}R_1$

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{3}{2} & -\frac{1}{2} & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$ $R_2 \to R_2 - 3R_1$, $R_3 \to R_3 - 2R_1$

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -1 & -1 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$ $R_2 \to 2R_2$

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & -1 & 0 \\ 0 & 1 & -1 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$ $R_1 \to R_1 - \frac{1}{2}R_2$

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & -1 \\ 0 & 1 & 0 & -4 & 2 & 1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$ $R_1 \to R_1 - R_3$, $R_2 \to R_2 + R_3$

$\rightarrow (I_3 / A^{-1})$

$\therefore A^{-1} = \begin{pmatrix} 3 & -1 & -1 \\ -4 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$

(OR)

41, (b) Let $z = x + iy$

$\frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1} = \frac{(x-1)+iy}{(x+1)+iy}$

$= \frac{(x-1)+iy}{(x+1)+iy} \times \frac{(x+1)-iy}{(x+1)-iy}$

$= \frac{(x+1)(x-1)+y^2 + i(y(x+1) - y(x-1))}{(x+1)^2 + y^2}$

$= \left[\frac{x^2 - 1 + y^2}{(x+1)^2 + y^2}\right] + i\left[\frac{2y}{(x+1)^2 + y^2}\right]$

$\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2} \Rightarrow R.P = 0$

$\frac{x^2 + y^2 - 1}{(x+1)^2 + y^2} = 0$

$x^2 + y^2 - 1 = 0$

42, (a) Given $\frac{1}{3}$ is a root of the equation $\therefore$ 3 is also a root.

$\therefore (x-\frac{1}{3})(x-3)=0$

$(3x-1)(x-3)=0$

$3x^2-10x+3$ is a factor.

$6x^4-5x^3-38x^2-5x+6$

$= (3x^2-10x+3)(2x^2+px+2)$

Equating x terms

$-5 = -20+3p$

$3p = 15 \Rightarrow p=5$

$\therefore$ other factor is $2x^2+5x+2$

$\therefore 2x^2+5x+2=0$

$(2x+1)(x+2)=0$

$2x+1=0$ ; $x+2=0$

$2x=-1$ $x=-2$

$x=-\frac{1}{2}$

$\therefore S.S = \{\frac{1}{3}; 3; -\frac{1}{2}; -2\}$

(OR)

42, (b) $\tan^{-1}(2x)+\tan^{-1}(3x)=\frac{\pi}{4}$

$\tan^{-1}\left(\frac{5x}{1-6x^2}\right)=\frac{\pi}{4}$

$\frac{5x}{1-6x^2} = \tan\frac{\pi}{4}$

$= 1$

$5x = 1-6x^2$

$6x^2+5x-1=0$

$6x^2+6x-x-1=0$

$6x(x+1)-1(x+1)=0$

$(6x-1)(x+1)=0$

$6x-1=0$ ; $x+1=0$

$6x=1$ $x=-1$

$x=\frac{1}{6}$

$x=-1$ is not a root.

$\therefore x=\frac{1}{6}$

43, (a) Width = 26m = 2a (major axis)

$a=13$

Height = 5m = b

$c^2 = a^2-b^2$

$c^2 = 169-25$

$c^2 = 144$

$c=12$

Focal distance = 2c

$=2\times 12$

$=24$ m

Hence the two person should stay at the distance of 12m from the centre point of the foci of the ellipse

P-V

(OR)

43, (b) $\vec{AB} = \vec{OB}-\vec{OA} = 10\vec{i}-12\vec{j}-4\vec{k}$.

$\vec{AC} = \vec{OC}-\vec{OA} = -6\vec{i}+10\vec{j}-6\vec{k}$.

$\vec{AD} = \vec{OD}-\vec{OA} = -4\vec{i}+2\vec{j}+10\vec{k}$.

$[\vec{AB}\ \vec{AC}\ \vec{AD}] = \begin{vmatrix} 10 & -12 & -4 \\ -6 & 10 & -6 \\ -4 & 2 & 10 \end{vmatrix} = 0$

$\therefore \vec{AB}, \vec{AC}, \vec{AD}$ are coplanar and $\therefore$ given points lie in a plane

44, (a) $L_1: \frac{x+1}{12} = \frac{y}{6} = \frac{z}{-1}$

$L_2: \frac{x}{6} = \frac{y+2}{6} = \frac{z-1}{1}$

Condition for skew lines

$(\vec{c}-\vec{a})\cdot(\vec{b}\times\vec{d}) \neq 0$.

$\vec{a} = -\vec{i}$

$\vec{c} = -2\vec{j}+\vec{k}$

$\vec{c}-\vec{a} = \vec{i}-2\vec{j}+\vec{k}$

$\vec{b} = 12\vec{i}+6\vec{j}-\vec{k}$

$\vec{d} = 6\vec{i}+6\vec{j}+\vec{k}$.

$(\vec{c}-\vec{a})\cdot(\vec{b}\times\vec{d}) = \begin{vmatrix} 1 & -2 & 1 \\ 12 & 6 & -1 \\ 6 & 6 & 1 \end{vmatrix}$

$= 12+36+36 = 84 \neq 0$

$\therefore$ the lines are skew.

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$$\vec{b}\times\vec{d} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 12 & 6 & -1 \\ 6 & 6 & 1 \end{vmatrix}$$

$$= 12\vec{i} - 18\vec{j} + 36\vec{k}$$

$$= 6(2\vec{i} - 3\vec{j} + 6\vec{k})$$

$$|\vec{b}\times\vec{d}| = 6\sqrt{4+9+36}$$

$$= 6\sqrt{49} = 6\times 7 = 42.$$

Distance between skew lines,

$$D = \frac{|(\vec{c}-\vec{a}).(\vec{b}\times\vec{d})|}{|\vec{b}\times\vec{d}|}$$

$$= \frac{84}{42} = 2 \text{ units}$$

(OR)

44. (b) $\frac{dv}{dt} = 1000\ cm^3$ ; $r = 7cm$

$\frac{dr}{dt} = ?$ $\frac{ds}{dt} = ?$

Volume $= V = \frac{4}{3}\pi r^3$

$$\frac{dv}{dt} = \frac{4}{3}\pi(3r^2)\frac{dr}{dt}$$

$$= 4\pi r^2 \frac{dr}{dt}$$

$$1000 = 4\pi(49)\frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1000}{4\pi\times 49} = \frac{250}{49\pi}$$

Surface area $= S = 4\pi r^2$

$$\frac{ds}{dt} = 4\pi\left(2r\frac{dr}{dt}\right)$$

$$= 8\pi r\frac{dr}{dt}$$

$$= 8\pi\times(7)\times\frac{250}{49\pi}$$

$$= \frac{250\times 8}{7}$$

rate of change of surface area $= \frac{2000}{7}\ cm^2$

(OR)

45. (c) $\lim_{n\to\infty} A = \lim_{n\to\infty} A_0\left(1+\frac{r}{n}\right)^{nt}$

$$= A_0\left[\lim_{n\to\infty}\left(1+\frac{r}{n}\right)^{nt}\right]$$

$\lim y = (1+\frac{r}{n})^{nt}$

$\log y = nt\log(1+\frac{r}{n})$

$$= \frac{\log(1+r/n)}{1/nt}$$

$\lim_{n\to\infty} \log y = \lim_{n\to\infty} \frac{\log 1}{0} = 0/0$

By L'H Rule

$$\lim_{n\to\infty} \log y = \lim_{n\to\infty} \frac{\frac{1}{1+\frac{r}{n}}\,r\left(-1/n^2\right)}{\frac{1}{t}\left(-1/n^2\right)}$$

$$= \lim_{n\to\infty} rt\times\frac{1}{1+\frac{r}{n}}$$

$$= rt\times\frac{1}{1+0}$$

$\log_e y = rt$

$y = e^{rt}$

$\therefore A = A_0(1+\frac{r}{n})^{nt}$

$= A_0 y$

$A = A_0 e^{rt}$

(OR)

45. (b) $u = \sec^{-1}\left(\frac{x^3-y^3}{x+y}\right)$

$$\sec u = \frac{x^3-y^3}{x+y} = f(x,y)$$

$$f(tx,ty) = \frac{t^3x^3 - t^3y^3}{tx+ty}$$

$$= \frac{t^3}{t}\left(\frac{x^3-y^3}{x+y}\right)$$

$$= t^2\left(\frac{x^3-y^3}{x+y}\right)$$

$$= t^2 f(x,y)$$

Degree $= 2$.

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$x\frac{\partial}{\partial x}(\sec u)+y\frac{\partial}{\partial y}(\sec u)=2\sec u$ (or) (b)

$x(\sec u\tan u)\frac{\partial u}{\partial x}+y(\sec u\tan u)\frac{\partial u}{\partial y}$

$=2\sec u$

$\div \sec u\tan u$

$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=2\times\frac{1}{\tan u}$

$=2\cot u$

| p | q | r | ~p | q∧r | p∨(q∧r) | (p∨q) | (p∨r) | (p∨q)∧(p∨r) |
|---|---|---|---|---|---|---|---|---|
| T | T | T | F | T | T | T | T | T |
| T | T | F | F | F | T | T | T | T |
| T | F | T | F | F | T | T | T | T |
| T | F | F | F | F | T | T | T | T |
| F | T | T | T | T | T | T | T | T |
| F | T | F | T | F | F | T | F | F |
| F | F | T | T | F | F | F | T | F |
| F | F | F | T | F | F | F | F | F |

$\therefore p\to(q\vee r)\equiv(p\wedge q)\vee(p\wedge r)$.

46 (a) $y=(x-2)^2+1$

$(x-2)^2=y-1$

vertex $=P(2,1)$

Let $Q(x,y)$ is another point.

Slope of $PQ=2$

$\frac{y-1}{x-2}=2\Rightarrow y-1=2x-4$

$2x-y-3=0$

$\Rightarrow y=2x-3$.

$y=(x-2)^2+1$

$\therefore (x-2)^2+1=2x-3$

$x^2-4x+4+1=2x-3$

$x^2-6x+8=0$

$(x-2)(x-4)=0$

$x=2,4$.

Limit $x=2$ to $x=4$

$A=\int_{x=2}^{x=4}(y_1-y_2)\,dx$

$=\int_2^4[(2x-3)-((x-2)^2+1)]\,dx$

$=\int_2^4[2x-3-(x^2-4x+4+1)]\,dx$

$=\int_2^4(2x-3-x^2+4x-5)\,dx$

$=\int_2^4(6x-x^2-8)\,dx$.

$=\left[3x^2-\frac{x^3}{3}-8x\right]_2^4$

$=\left(3(16)-\frac{64}{3}-32\right)-\left(12-\frac{8}{3}-16\right)$

47 (a) $\frac{dv}{dt}=g\cos\alpha-kv$

$\frac{dv}{g\cos\alpha-kv}=dt$

$\frac{-k\,dv}{g\cos\alpha-kv}=-k\,dt$

$\int\frac{-k\,dv}{g\cos\alpha-kv}=-k\int dt$

$\log_e(g\cos\alpha-kv)=-kt+c$

$g\cos\alpha-kv=e^{-kt+c}$

$=e^{-kt}e^c$

$=ce^{-kt}$

when $t=0$, $v=0$

$g\cos\alpha=c(1)$

$=c$

$g\cos\alpha-kv=g\cos\alpha\,e^{-kt}$

$g\cos\alpha(1-e^{-kt})=kv$.

$v=\frac{g\cos\alpha}{k}(1-e^{-kt})$

(OR)

47 (b) Let $x$ is the number of heads.

$x$ takes the values $0,1,2,3,4$

$p=\frac{1}{2}\quad q=\frac{1}{2}$

$P(x=0)=\left(\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\right)=\frac{1}{16}$

$P(x=1)=4\left(\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\right)=\frac{4}{16}$

$P(x=2)=6\left(\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\right)=\frac{6}{16}$

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| $X$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X=x)$ | $\frac{1}{16}$ | $\frac{4}{16}$ | $\frac{6}{16}$ | $\frac{4}{16}$ | $\frac{1}{16}$ |

Here $n = 4$, $p = \frac{1}{2}$ $q = \frac{1}{2}$

mean $= np = 4 \times \frac{1}{2} = 2$

Variance $= npq = 4 \times \frac{1}{2} \times \frac{1}{2} = 1$

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